

Active Inference Symposium

Money & The Free Energy Principle

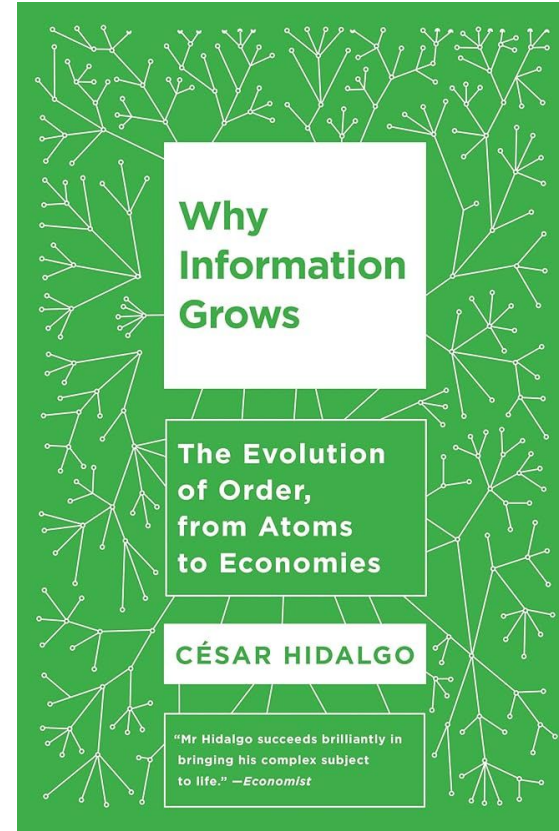
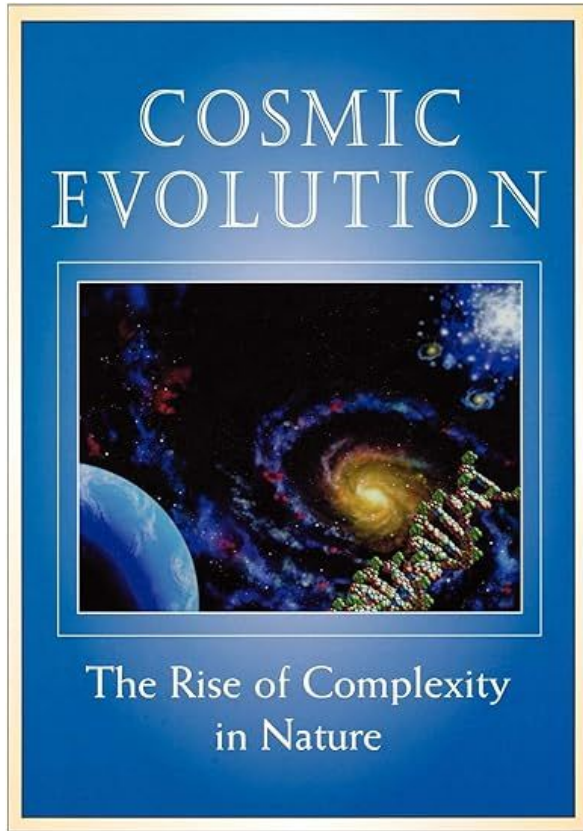
Steph Macurdy, Head of Blockchain Research, Wolfram
Andrius Kulikauskas, PhD, Math4Wisdom

WOLFRAM
BLOCKCHAIN LABS

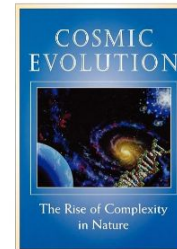


Presentation

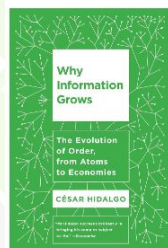
1. My Story
2. Proof of Entropy Minimization
3. Math of Energy, Entropy & Free Energy



Free Energy Maximization

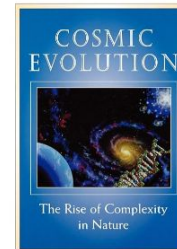


- 2015
- 2017
- 2020
- 2024



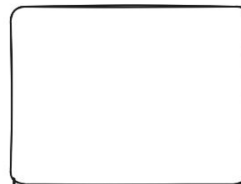
Entropy Minimization

Free Energy Maximization



Energy
Computation

Self-Organization



2015

2017

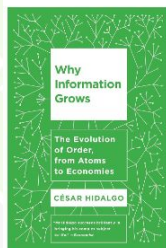
2020

2024

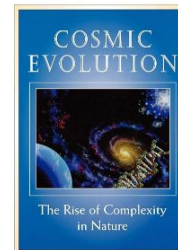
Proof of Work

Distributed
Consensus

Entropy
Minimization



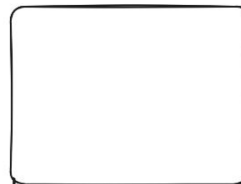
Free Energy Maximization



Energy
Computation

Self-Organization

Free Energy
(sunlight)

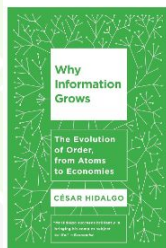
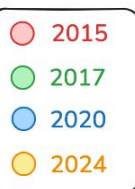


Free Energy
Density (batteries)

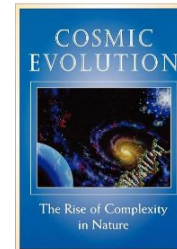
Distributed
Consensus

Proof of Work

Entropy
Minimization



Free Energy Maximization



Energy
Computation

Self-Organization

Emergent
Complexity

Discrete
Space

Free Energy
(sunlight)

Computation

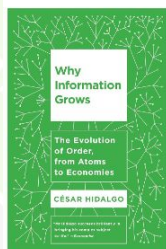
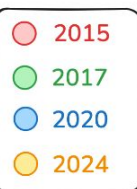
Observer
Theory

Free Energy
Density (batteries)

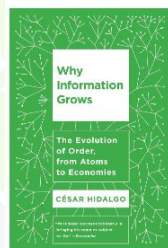
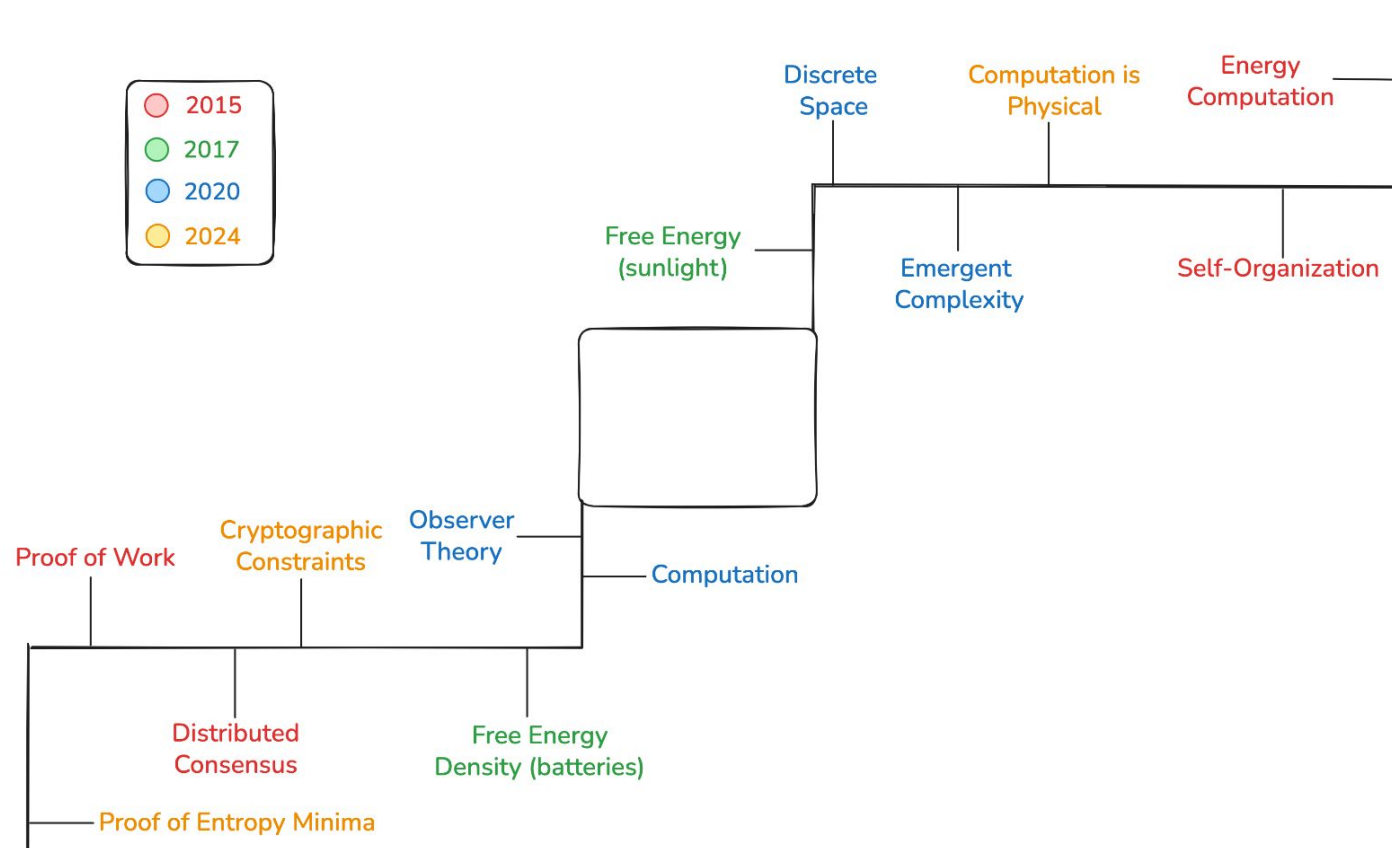
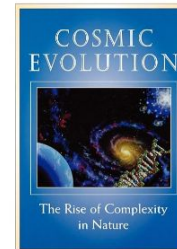
Distributed
Consensus

Proof of Work

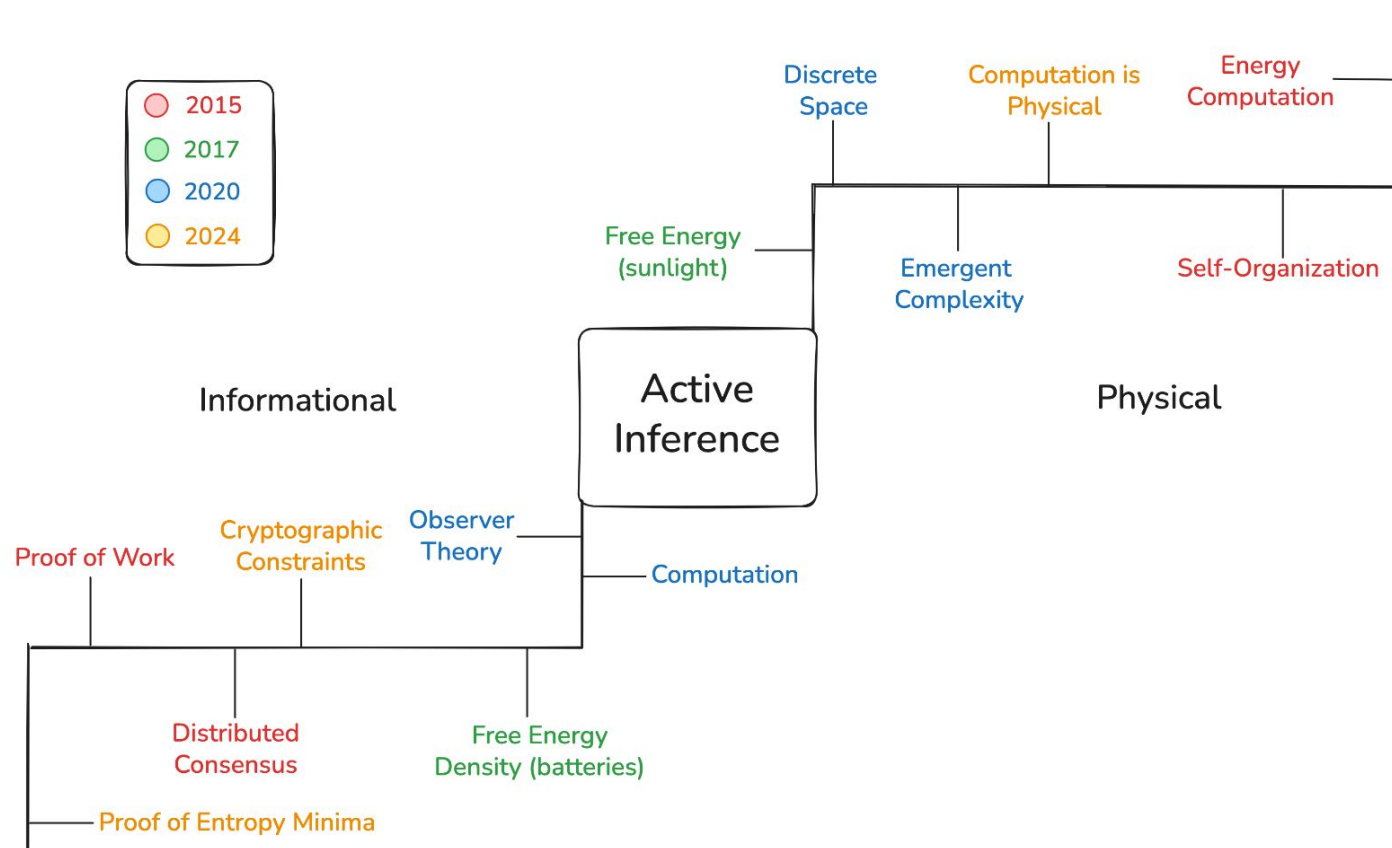
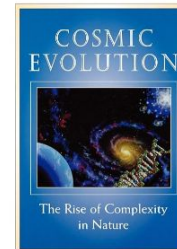
Entropy
Minimization



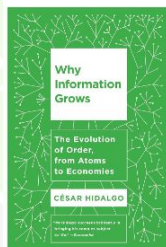
Free Energy Maximization



Free Energy Maximization



Entropy Minimization



Variant of Proof of Work

POEM: Proof of Entropy Minima

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March 7, 2023

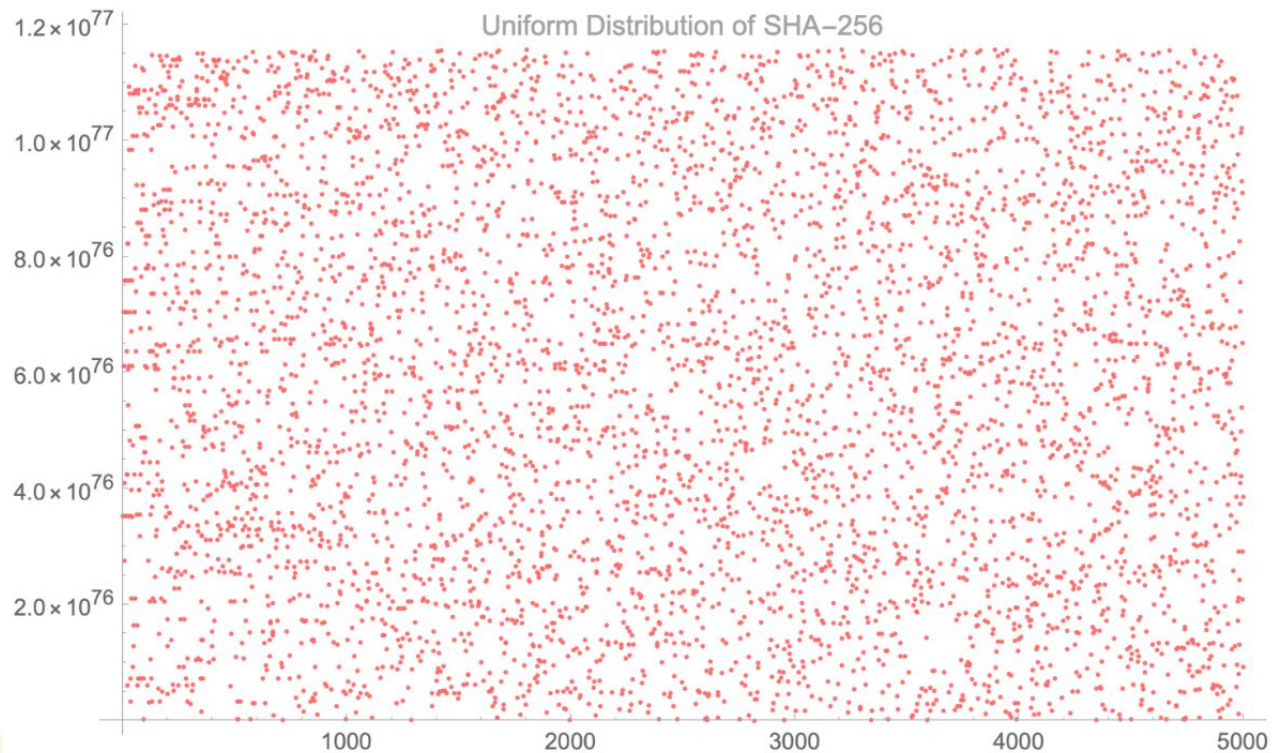
Abstract

Nakamoto consensus has been incredibly influential in enabling robust blockchain systems, and one of its components is the so-called heaviest chain rule (HCR). Within this rule, the calculation of the weight of the chain tip is performed by adding the difficulty threshold value to the previous total difficulty. Current difficulty based weighting systems do not take the intrinsic block weight into account. This paper proposes a new mechanism based on entropy differences, named proof of entropy minima (POEM), which incorporates the intrinsic block weight in a manner that significantly reduces the orphan rate of the blockchain while simultaneously accelerating finalization. Finally, POEM helps to understand blockchain as a static time-independent sequence of committed events.

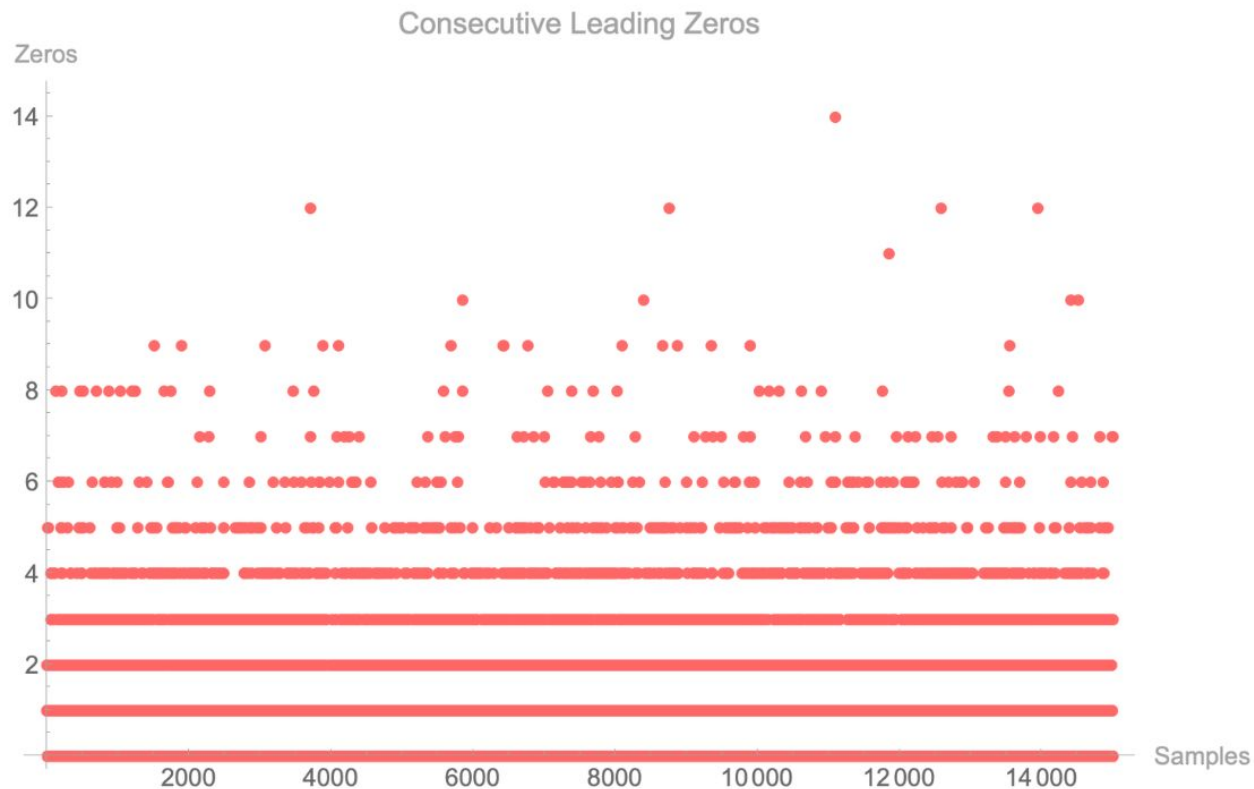
Proof of (Shannon) Entropy Minimization

- The total possible state space of a SHA-256 output = 2^{256}
- Rule: # of consecutive leading 0s in SHA-256 output (beyond arbitrary threshold)
- Example:
 - 0000000010110100111010101110011011011010110000100000010010111111111110000011
000010101001010110101001010110011001000111001001111010000010010110101010011001
011001010111001011011010100110001010001011100101010001110011110
- Interpretation: # of bits of uncertainty removed (accumulated/minimized)

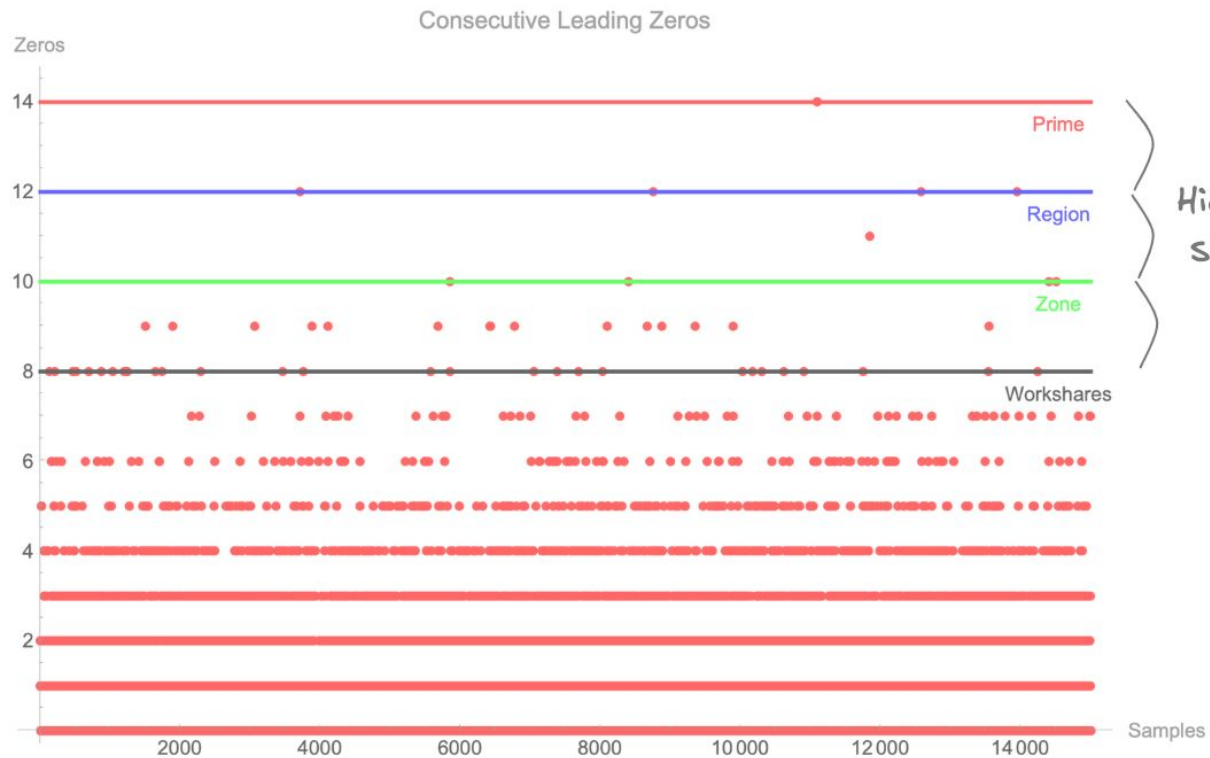
Maximum Entropy



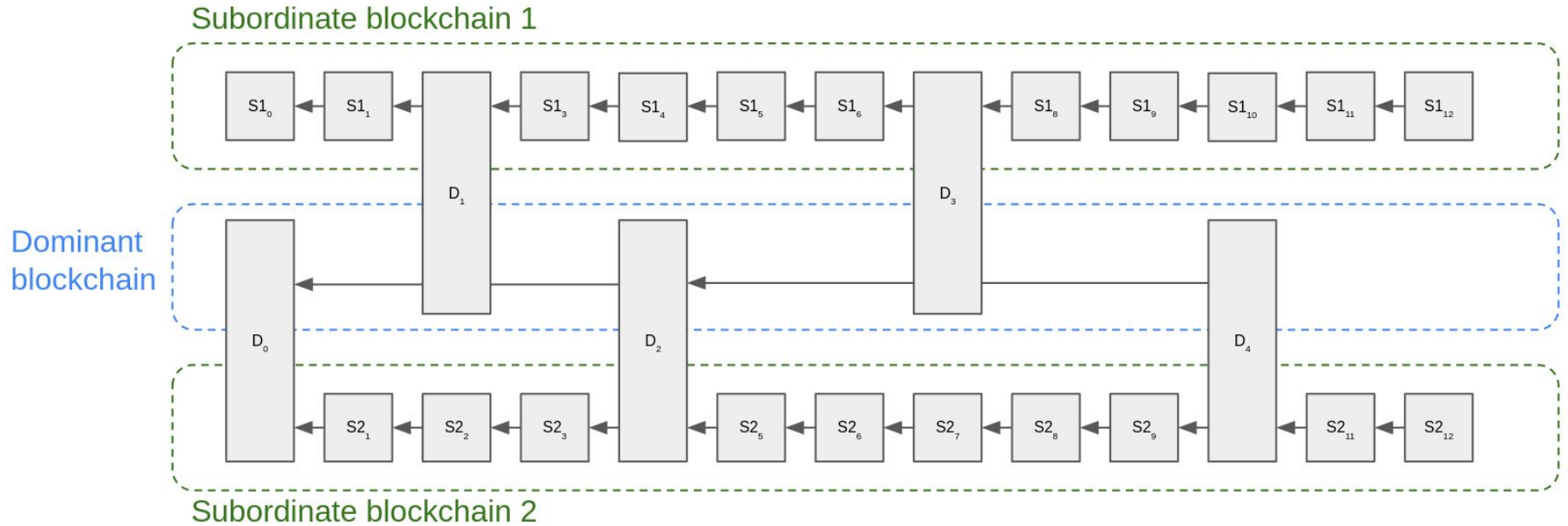
Maximum Entropy + Constraint



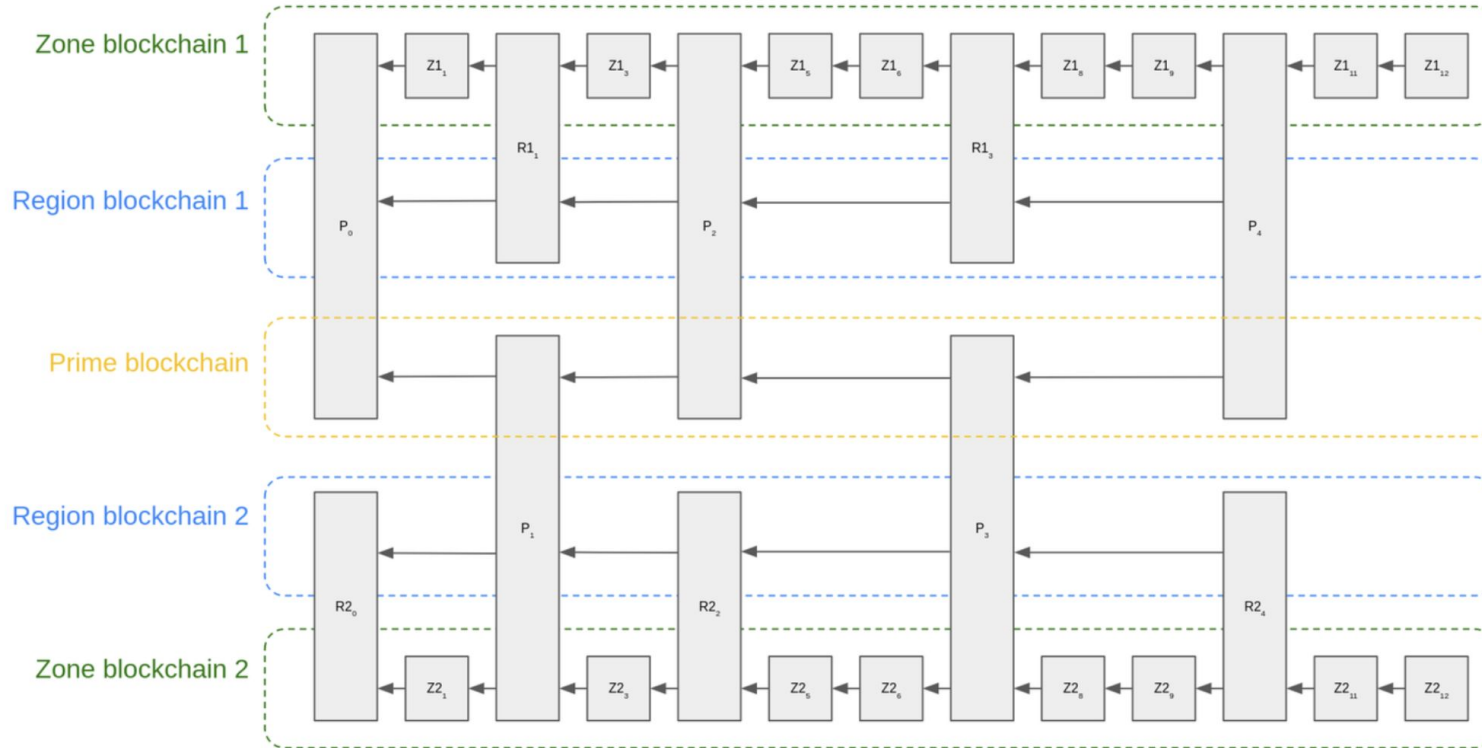
Hierarchical Sequences



Hierarchical Blockchains



Quai Network



Simple Rule, Emergent Results

Proof of Entropy Minima

$$\triangle S = 1/2^n$$

of states removed

SHA256



```
0000000000000000
0000000000000000
0000110001100
111010101100.
```

"A difference that makes a difference."

The highlighted red zero is a crucial, deterministic factor in the measurement of work (via statistical entropy) for blockchain consensus mechanisms. This single bit, which is forfeited in traditional PoW (longest chain rules) is employed in PoEM (heaviest chain rules), and is the difference between achieving instantaneous finalization and waiting for multiple rounds of validation. It is the difference between excess thermodynamic waste, and precise measurement of free energy expenditure. It is intrinsic to the system itself, which resolves two valid competitive outputs without conflict. This means that consensus can be established without contention, reaching finality as fast as physically possible. This forms the basis of sharded (scalable) proof of work system and the characteristic known as "causal invariance."

Quai Network

\$QUAI

- **Store of Value**
- **Free Energy**

\$QI

- **Medium of Exchange**
- **Unit of Account**
- **Entropy**

Part 1

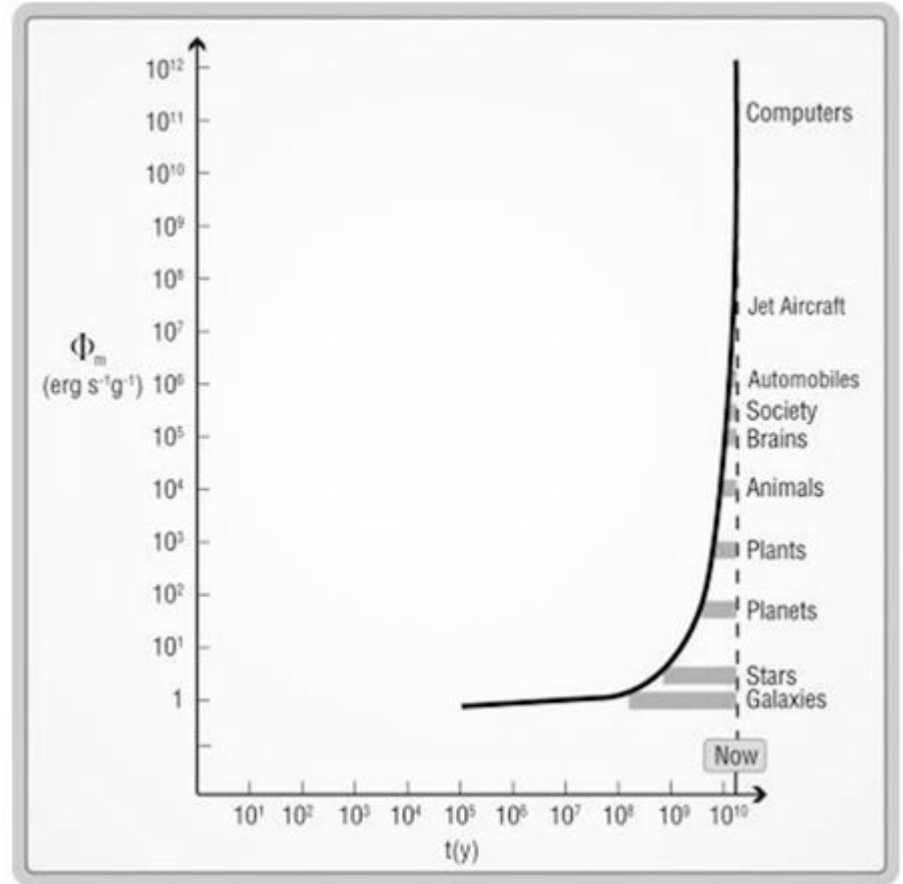
How much [does] a dollar cost?

High Road

Free Energy Rate Density

$$\Phi_m$$

(ergs/second/gram)



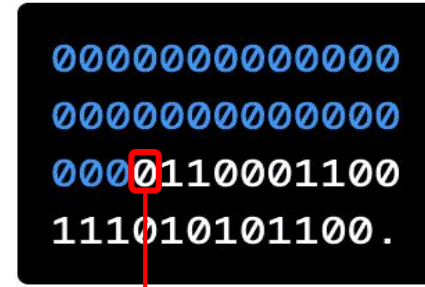
Low Road

Proof of Entropy Minima

$$\triangle S = 1/2^n$$

of states removed

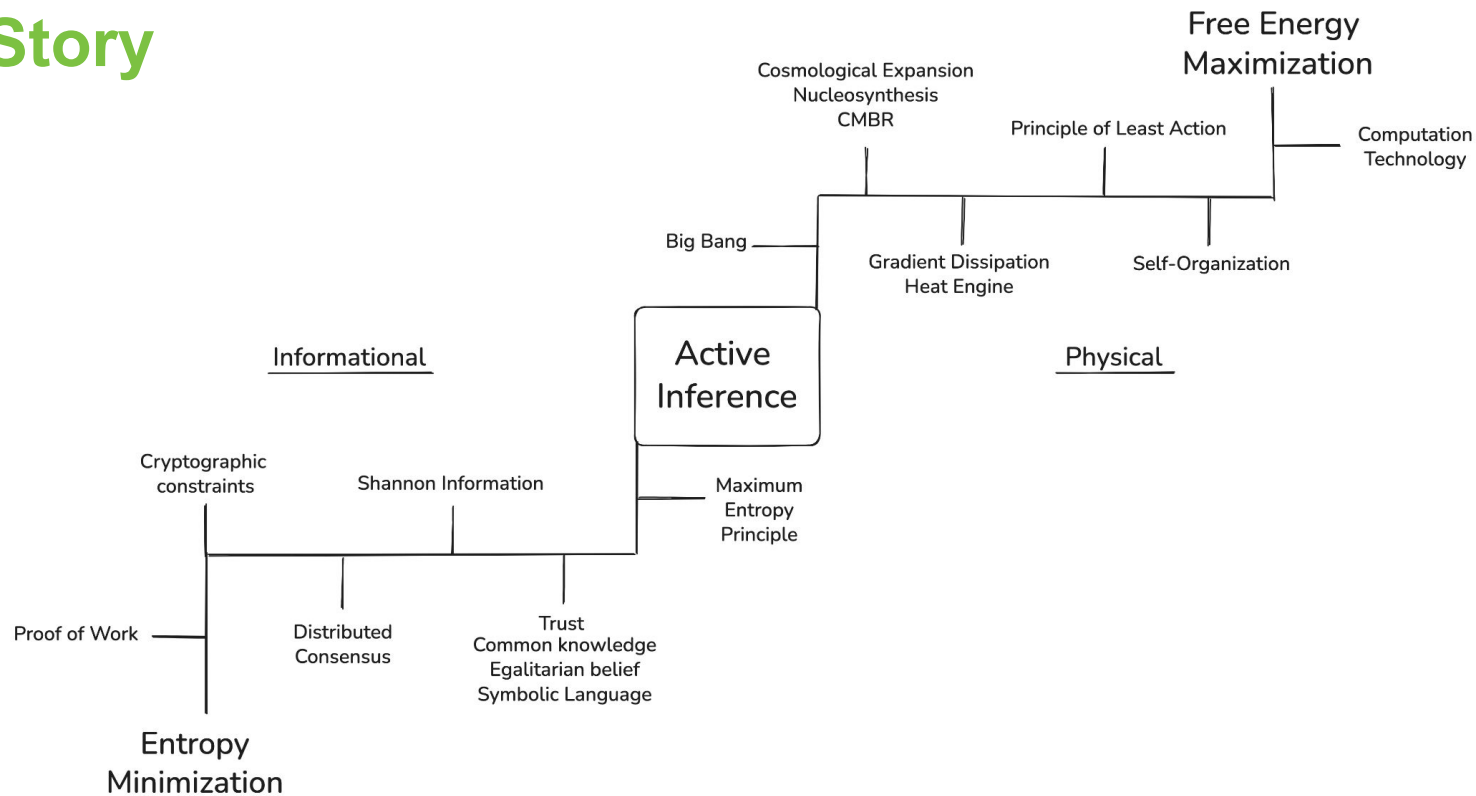
SHA256



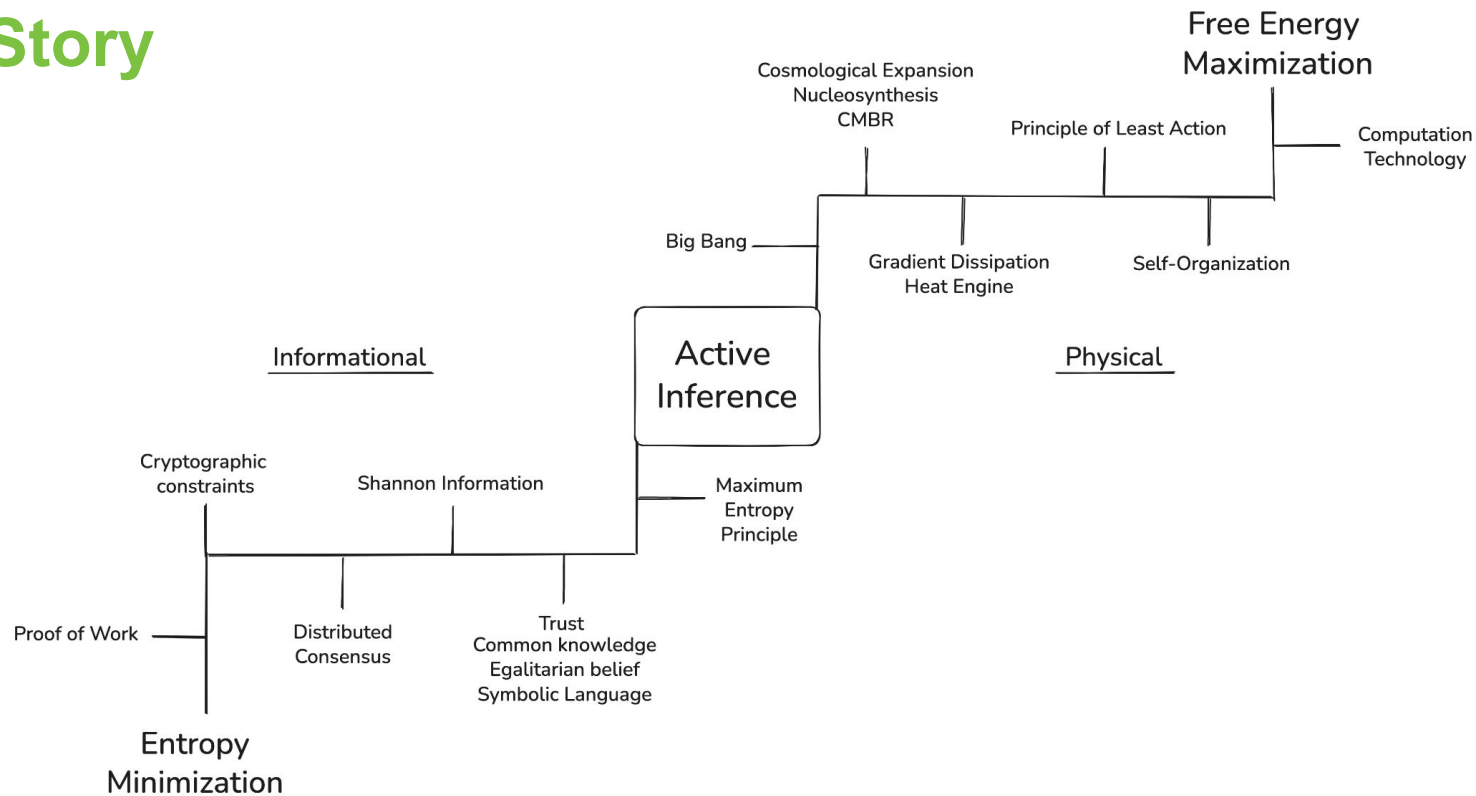
"A difference that makes a difference."

The highlighted red zero is a crucial, deterministic factor in the measurement of work (via statistical entropy) for blockchain consensus mechanisms. This single bit, which is forfeited in traditional PoW (longest chain rules) is employed in PoEM (heaviest chain rules), and is the difference between achieving instantaneous finalization and waiting for multiple rounds of validation. It is the difference between excess thermodynamic waste, and precise measurement of free energy expenditure. It is intrinsic to the system itself, which resolves two valid competitive outputs without conflict. This means that consensus can be established without contention, reaching finality as fast as physically possible. This forms the basis of sharded (scalable) proof of work system and the characteristic known as "causal invariance."

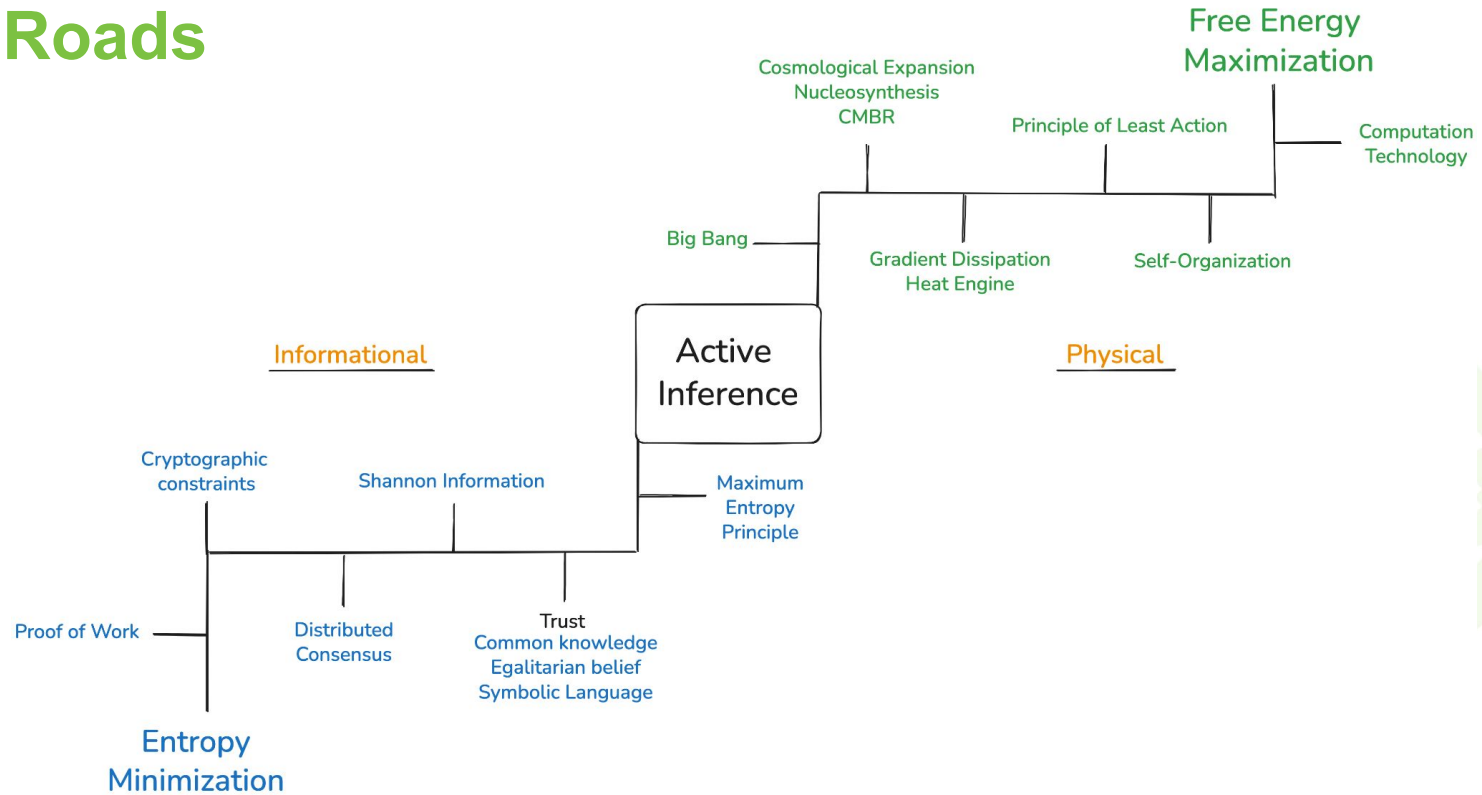
My Story



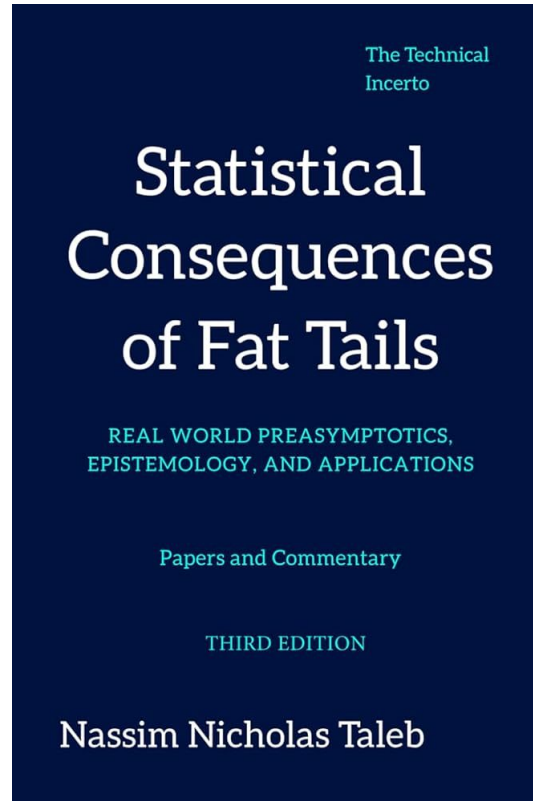
My Story



Two Roads



Risk-Adjusted Return



The diagram illustrates the formula for calculating the total number of blocks in Bitcoin, accounting for halving events. The formula is:

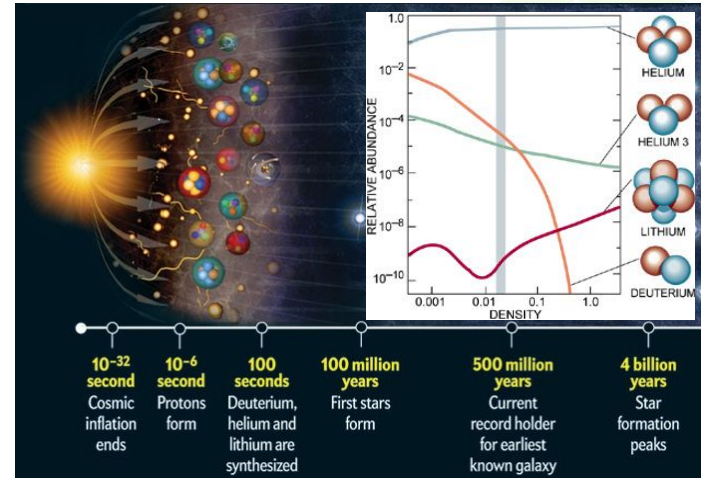
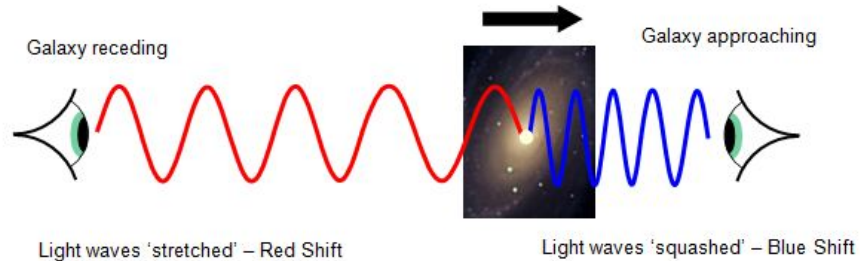
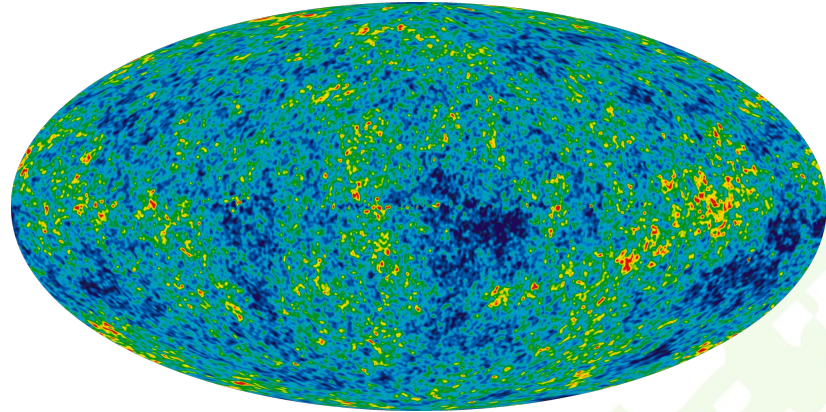
$$\sum_{i=0}^{32} 210,000 * \left\lfloor \frac{50}{2^i} \right\rfloor$$

The components of the formula are annotated as follows:

- 32**: The total number of halving epochs
- $\sum$** : The symbol for a summation equation
- $i=0$** : Initial Epoch
- 210,000**: The # of blocks in each halving epoch
- $*$** : The Halving
- 50**: The maximum (initial) block reward
- 2^i** : The current halving epoch

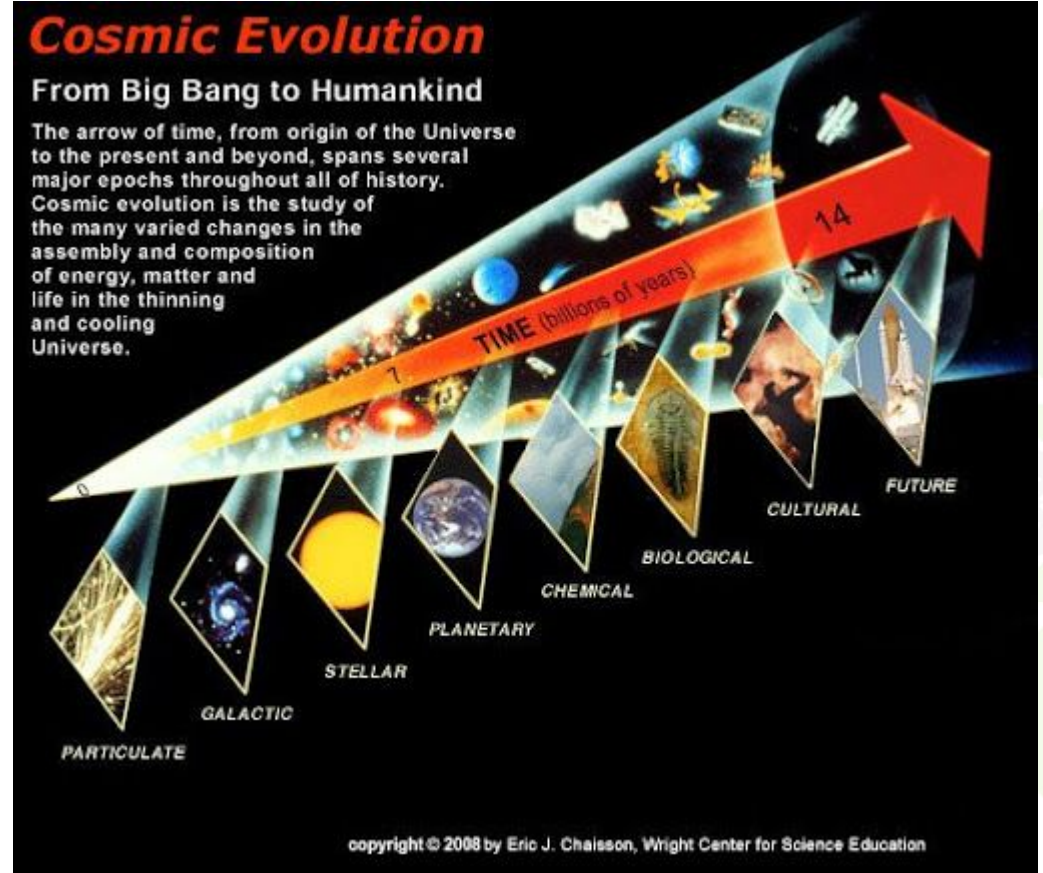
Big Bang Evidence

1. Cosmic Microwave Background Radiation
2. Expansion observed by Doppler effect
3. Nucleosynthesis



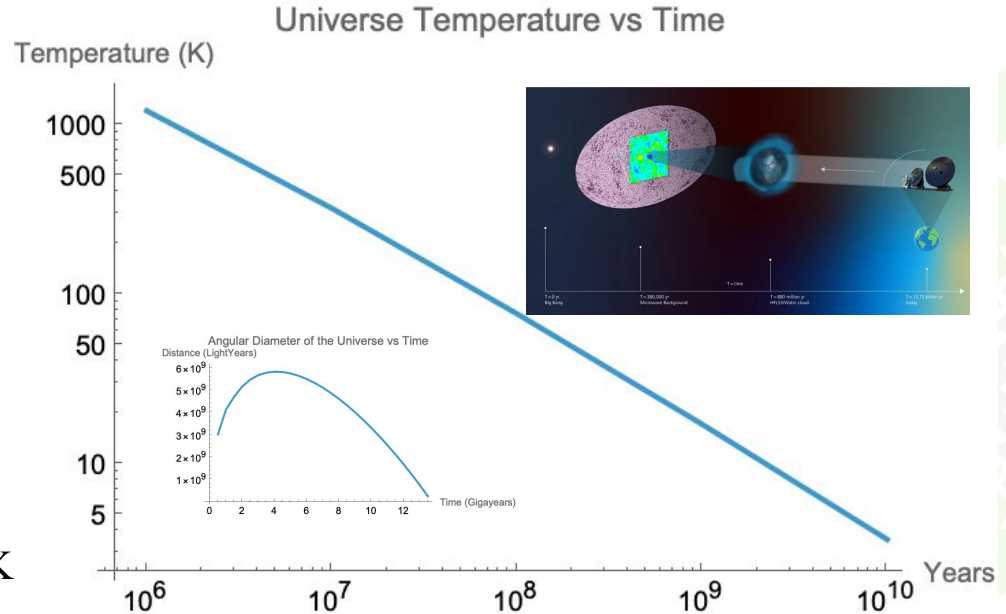
Epochs of Evolution

- Energetic
- Atomic
- Galactic
- Stellar
- Planetary
- Chemical
- Biological
- Cultural, Technological, Computational, Symbolic.

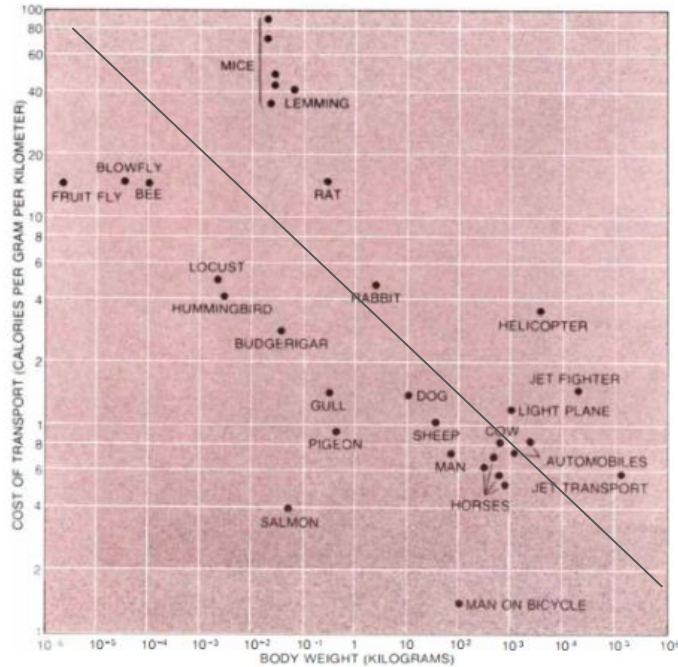


The Universe is a Temperature Gradient (Heat Engine)

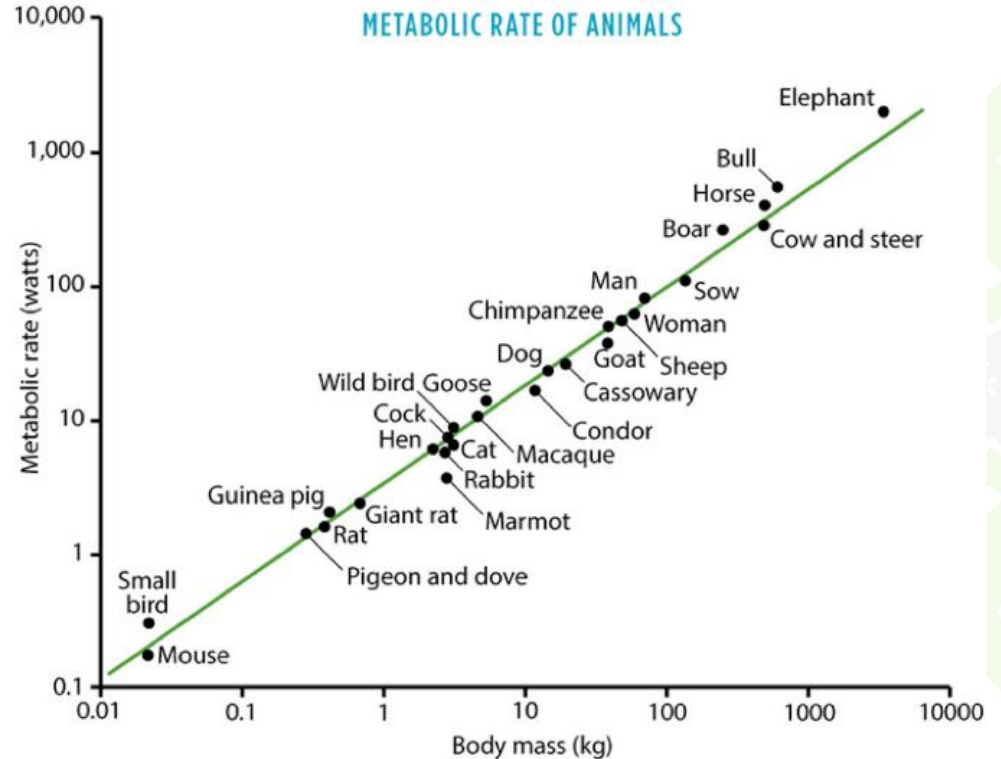
- 13.8 billion years, $T = 2.7255$ K,
- 10 billion years, $T = 3.62$ K,
- 7 billion years, $T = 4.77$ K,
- 1 billion years, $T = 18.12$ K,
- 100 million years, $T = 83.4$ K,
- 10 million years, $T = 379$ K,
- 1 million years, $T = 1633$ K,
- 25,000 years, $T = 13,700$ K,
- 1000 years, $T = 73,600$ K,
- 1 year, $T = 2,370,000$ K,
- 1 second, $T = 10,000,000,000$ K,
- 1 microsecond, $T = 10,000,000,000,000$ K



Metabolic Gradient



MAN ON A BICYCLE ranks first in efficiency among traveling animals and machines in terms of energy consumed in moving a certain distance as a function of body weight. The rate of energy consumption for a bicyclist (about .15 calorie per gram per kilometer) is approximately a fifth of that for an unaided walking man (about .75 calorie per gram per kilometer). With the exception of the black point representing the bicyclist (lower right), this graph is based on data originally compiled by Vance A. Tucker of Duke University.



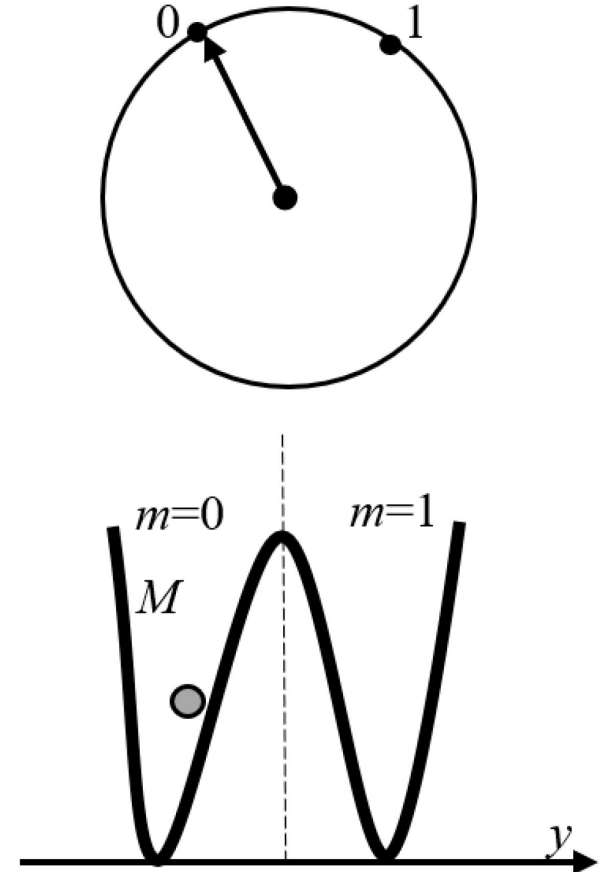
Part 2

The Descent

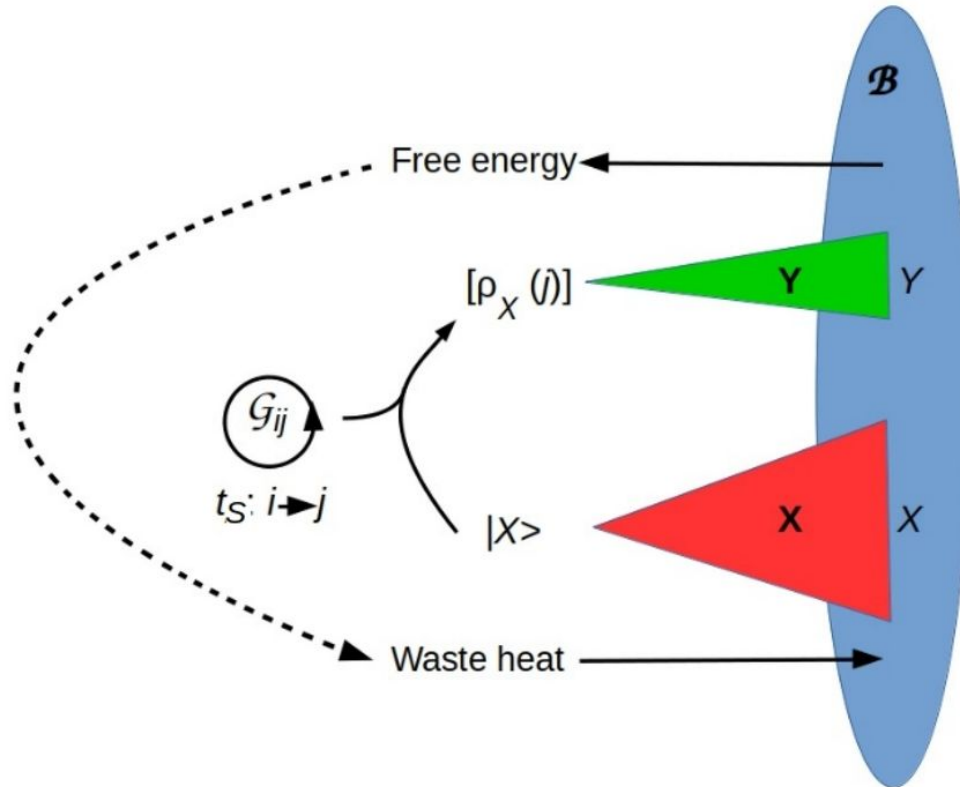
Information is Physical

“Landauer’s Bound”

Exemplification of the Landauer principle is supplied by the Brownian particle in a double-well potential. When the barrier is much higher than the thermal energy, the particle will remain in either well for a long time. Thus, the particle being in the left or right well can serve as the stable informational states, “0” and “1” of a bit. A Brownian particle trapped in either left or right well represents the informational states $W = k_B T \ln 2$. The experimental verification of the Landauer principle was carried out with an optical tweezer, which trapped a silica bead (2 μm in diameter) at the focus of a laser beam. It was demonstrated that the mean dissipated heat saturated at the Landauer bound in the limit of long erasure cycles. The extension of the Landauer principle to the quantum realm is carried out by using a crystal of molecular nanomagnets as a quantum spin memory.



Computation is Work



Representation of a system A that deploys a Quantum Reference Frame (QRF) X (red triangle) to measure the state of an external system X in its informational environment (i.e., a sector X of its boundary $\mathcal{B}$), and then, deploys a second QRF Y (green triangle) to write the outcome to a memory sector Y. This process induces one “tick” of an internal clock $\mathcal{G}_{ij}$ that defines an internal elapsed time t_S . The process is powered by a thermodynamic loop from (thermodynamic free energy in) and back to (waste heat out) the physical environment $\mathcal{E}$.

Principle of Least Action

$$\delta \int_{t_0}^{t_1} L(q, \dot{q}, t) dt = 0$$

Let's say a system moves from time t_1 to t_2 . The **action** S is defined as:

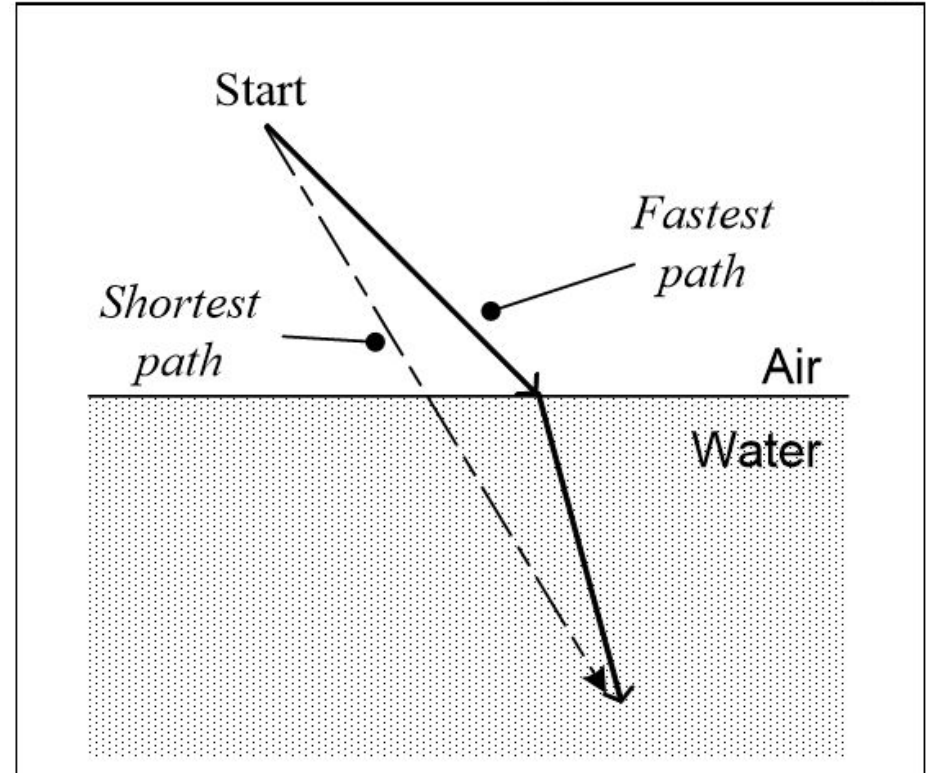
$$S = \int_{t_1}^{t_2} L(q, \dot{q}, t) dt$$

- L is the **Lagrangian** of the system: $L = T - V$
(kinetic energy minus potential energy)
- $q(t)$ are the generalized coordinates (e.g. position)
- $\dot{q}(t)$ are their time derivatives (e.g. velocity)

The Principle of Least Action says:

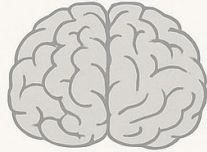
$$\delta S = 0$$

—meaning the actual path taken by the system makes the action stationary (minimum, maximum, or a saddle point).



Entropy

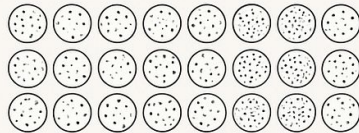
Thermodynamic Entropy



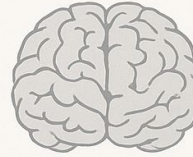
Physical Capacity

$$S = k \log \Omega$$

- Based on physical states
- Maximum structure



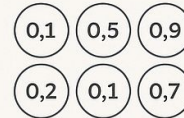
Shannon Entropy



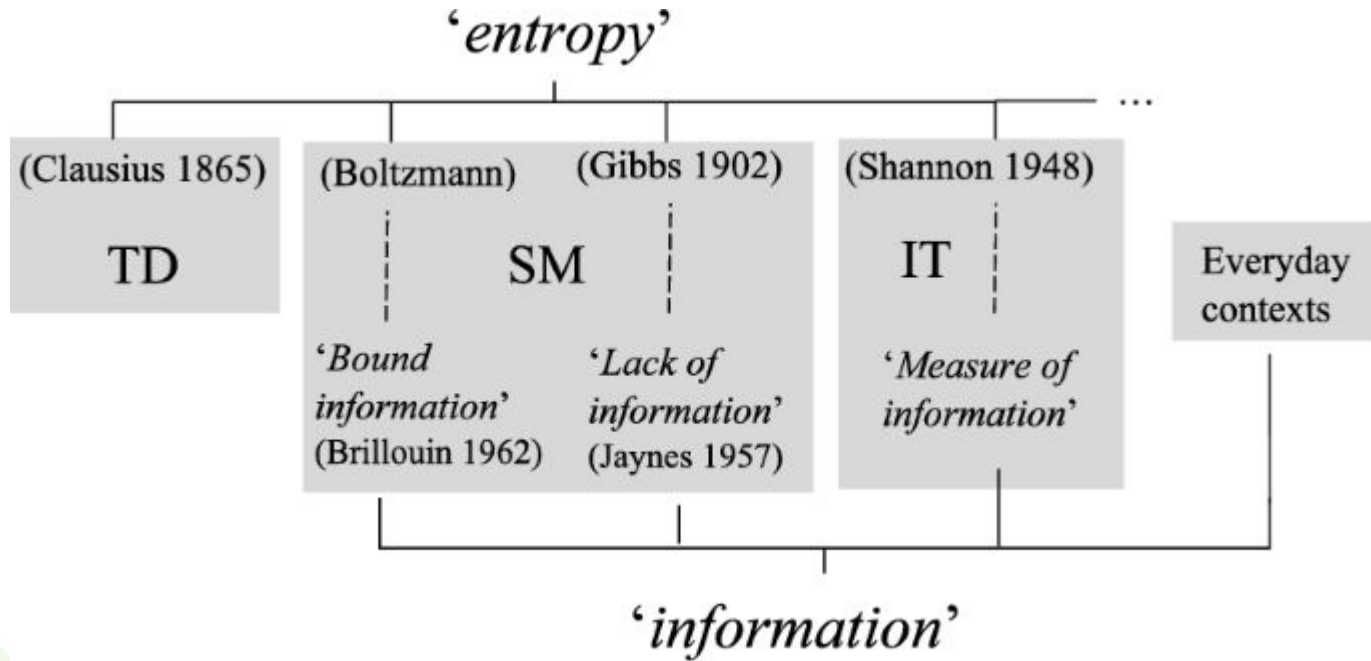
Informational Capacity

$$H = -\sum p_i \log_2 p_i$$

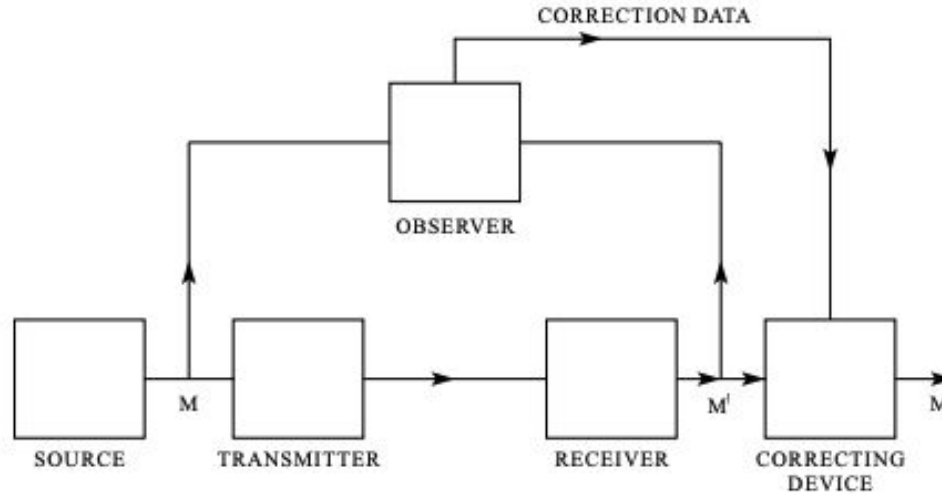
- Based on probabilities
- Maximum information



Entropy



“Mathematical Theory of Communication.” Shannon



Theorem 17: The capacity of a channel of band W perturbed by white thermal noise power N when the average transmitter power is limited to P is given by

$$C = W \log_2 \frac{P+N}{N}.$$

This means that by sufficiently involved encoding systems we can transmit binary digits at the rate $W \log_2 \frac{P+N}{N}$ bits per second, with arbitrarily small frequency of errors. It is not possible to transmit at a higher rate by any encoding system without a definite positive frequency of errors.

Work + Information have the same formalism

Information Theory and Statistical Mechanics

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(Received September 4, 1956; revised manuscript received March 4, 1957)

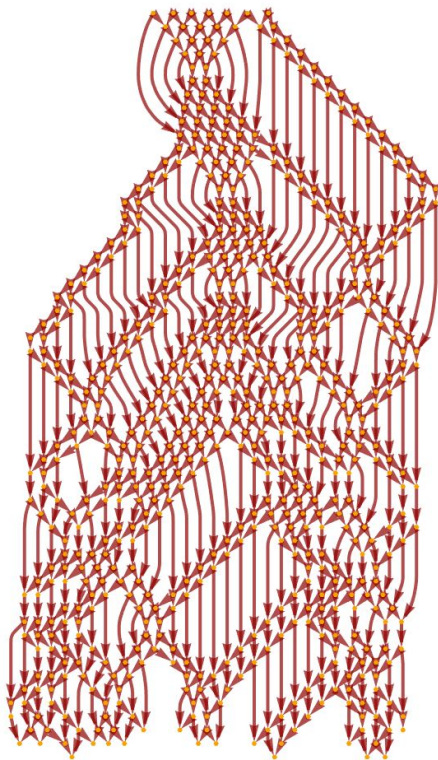
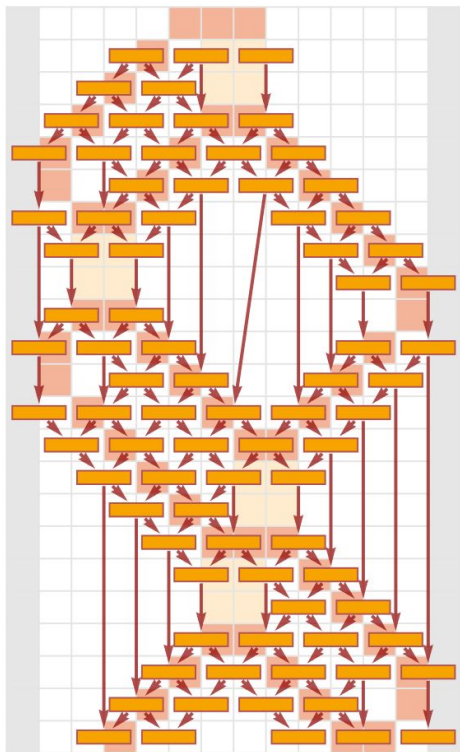
Information theory provides a constructive criterion for setting up probability distributions on the basis of partial knowledge, and leads to a type of statistical inference which is called the maximum-entropy estimate. It is the least biased estimate possible on the given information; i.e., it is maximally noncommittal with regard to missing information. If one considers statistical mechanics as a form of statistical inference rather than as a physical theory, it is found that the usual computational rules, starting with the determination of the partition function, are an immediate consequence of the maximum-entropy principle.

In the resulting "subjective statistical mechanics," the usual rules are thus justified independently of any physical argument, and in particular independently of experimental verification; whether

or not the results agree with experiment, they still represent the best estimates that could have been made on the basis of the information available.

It is concluded that statistical mechanics need not be regarded as a physical theory dependent for its validity on the truth of additional assumptions not contained in the laws of mechanics (such as ergodicity, metric transitivity, equal *a priori* probabilities, etc.). Furthermore, it is possible to maintain a sharp distinction between its physical and statistical aspects. The former consists only of the correct enumeration of the states of a system and their properties; the latter is a straightforward example of statistical inference.

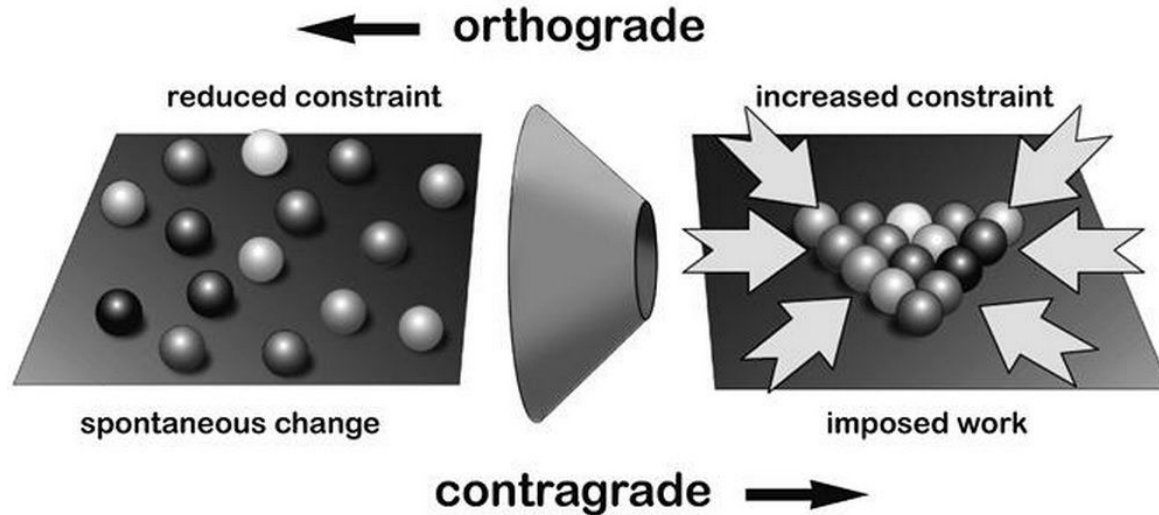
Computational Irreducibility



Which pathway?

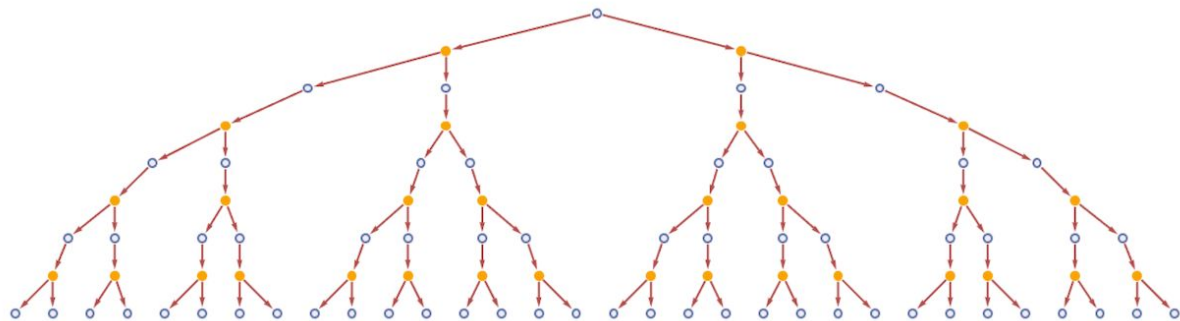
- Universe is discrete.
- Persistent entropy associated with observer-dependent, coarse-grained perception of spacetime.
- Increased “organization” of matter could be more than balanced by enlargement in the number of available states for spacetime.

Constraint: “A difference that makes a difference.”

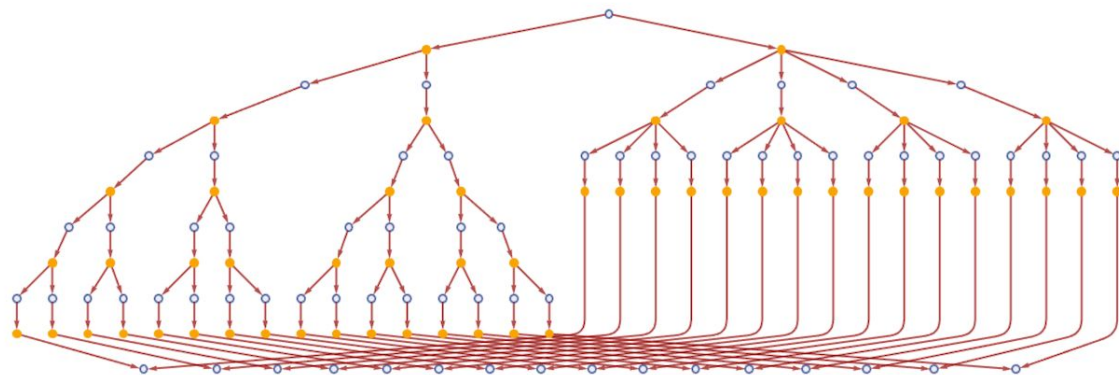


A cartoon characterization of the asymmetry implicit in thermodynamic change from a constrained (“ordered”) state to a less constrained (“disordered”) state, which tends to occur spontaneously (an orthograde process), contrasted with the reversed direction of change, which does not tend to occur spontaneously (a contragrade process), and so only tends to occur in response to the imposition of highly constrained external work (arrows in the image on the right).

Causal Invariance vs Confluence



Isomorphic; therefore
causally invariant



Non-isomorphic; not
causally invariant

Free Energy Principle

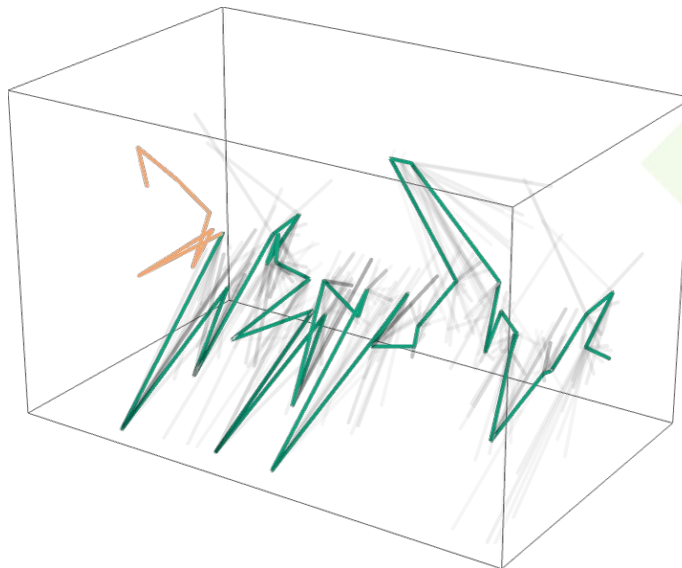
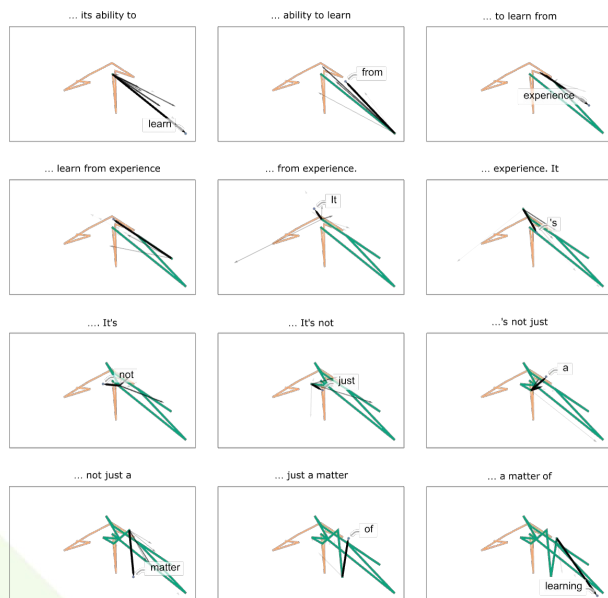
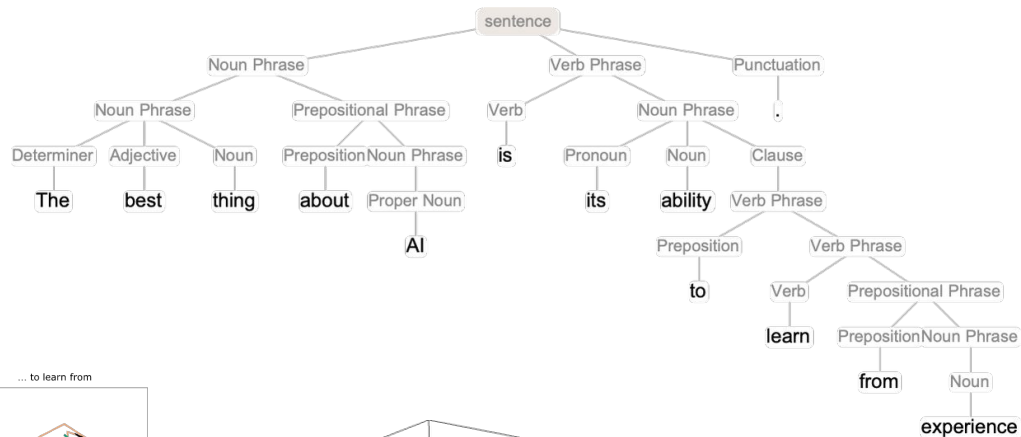
Free energy

Surprise

$$\mathcal{F}(y, \zeta) = D_{KL}(q(x; \zeta) || p(x|y)) - \ln(p(y)) \geq -\ln(p(y))$$

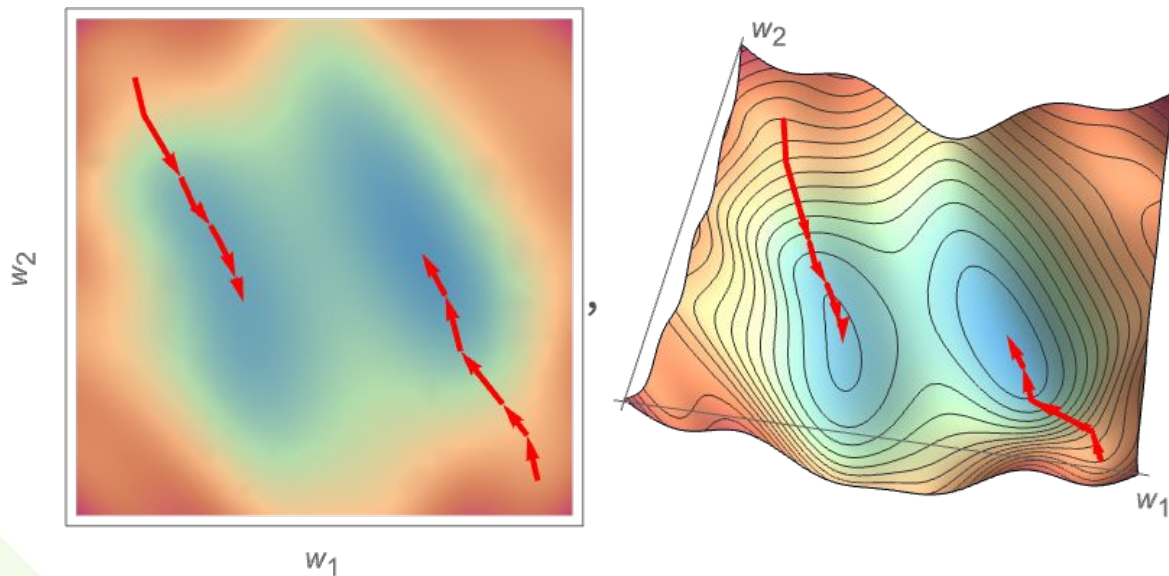
$$\begin{aligned} G(\pi) &= - \underbrace{\mathbb{E}_{Q(\tilde{x}, \tilde{y}|\pi)}[D_{KL}[Q(\tilde{x}|\tilde{y}, \pi) || Q(\tilde{x}|\pi)]]}_{\text{Information gain}} - \underbrace{\mathbb{E}_{Q(\tilde{y}|\pi)}[\ln P(\tilde{y}|C)]}_{\text{Pragmatic value}} \\ &= \underbrace{\mathbb{E}_{Q(\tilde{x}|\pi)}[H[P(\tilde{y}|\tilde{x})]]}_{\text{Expected ambiguity}} + \underbrace{D_{KL}[Q(\tilde{y}|\pi) || P(\tilde{y}|C)]}_{\text{Risk (outcomes)}} \\ &\leq \underbrace{\mathbb{E}_{Q(\tilde{x}|\pi)}[H[P(\tilde{y}|\tilde{x})]]}_{\text{Expected ambiguity}} + \underbrace{D_{KL}[Q(\tilde{x}|\pi) || P(\tilde{x}|C)]}_{\text{Risk (states)}} \\ &= - \underbrace{\mathbb{E}_{Q(\tilde{x}, \tilde{y}|\pi)}[\ln P(\tilde{y}, \tilde{x}|C)]}_{\text{Expected energy}} - \underbrace{H[Q(\tilde{x}|\pi)]}_{\text{Entropy}} \end{aligned}$$

Gradient Descent



Gradient Descent

Repeat until converge {
 $w = w - \alpha \left[\frac{\partial Loss}{\partial w} \right]$
 $b = b - \alpha \left[\frac{\partial Loss}{\partial b} \right]$
}

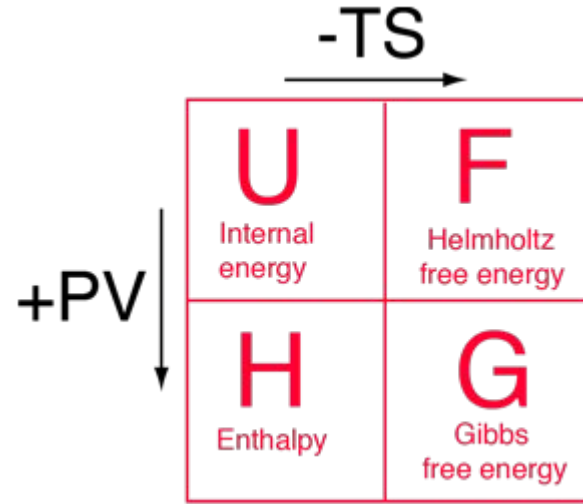


Part 3

Proof of Entropy Minima

Free Energy

Helmholtz and Gibbs free energies are both thermodynamic potentials that measure the amount of useful work obtainable from a system. Helmholtz free energy is used at constant temperature and volume, while Gibbs free energy is used at constant temperature and pressure



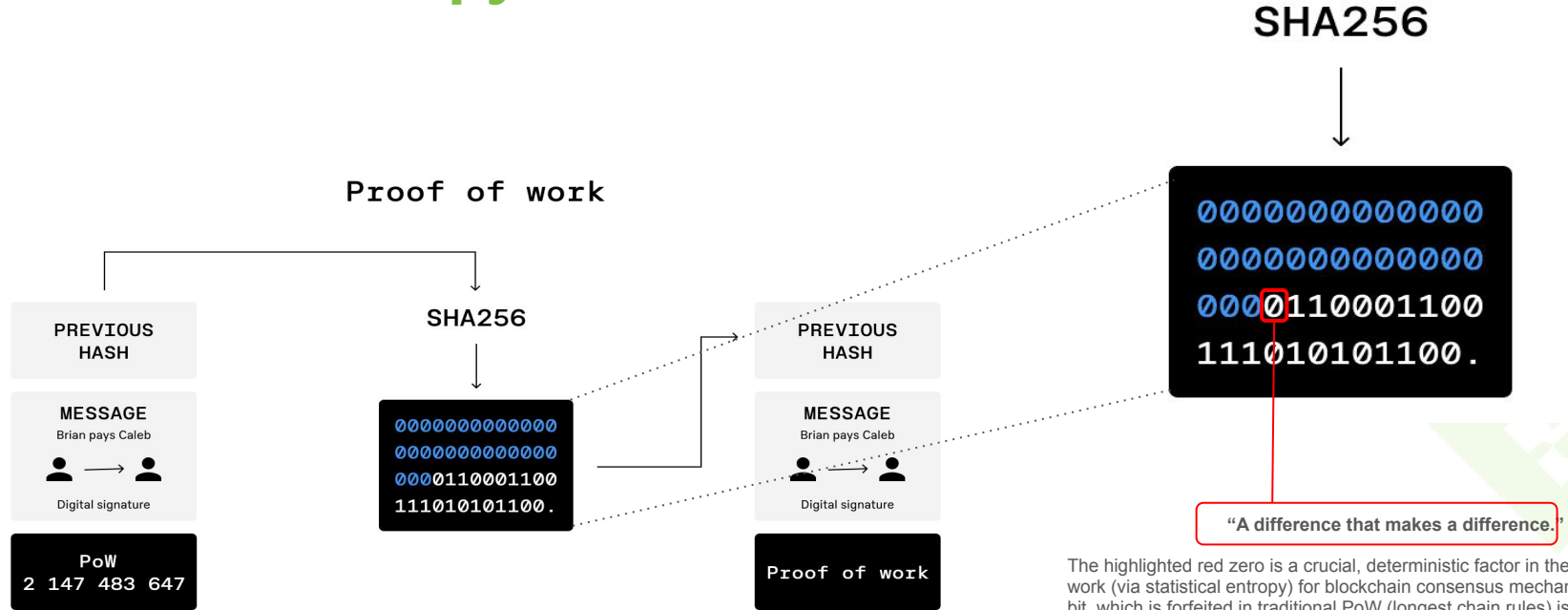
$$G = U - TS + PV$$

G Gibbs free energy **U** Internal energy **-TS** Final entropy **+PV** Final volume

Absolute temperature Absolute pressure

Energy you can get from the system's environment by heating Work to give the system final volume V at constant pressure P

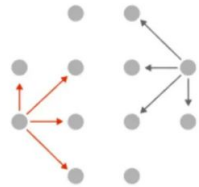
Proof of Entropy Minima



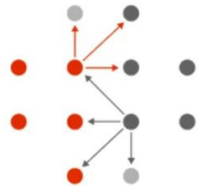
The highlighted red zero is a crucial, deterministic factor in the measurement of work (via statistical entropy) for blockchain consensus mechanisms. This single bit, which is forfeited in traditional PoW (longest chain rules) is employed in PoEM (heaviest chain rules), and is the difference between achieving instantaneous finalization and waiting for multiple rounds of validation. It is the difference between excess thermodynamic waste, and precise measurement of free energy expenditure. It is intrinsic to the system itself, which resolves two valid competitive outputs without conflict. This means that consensus can be established without contention, reaching finality as fast as physically possible. This forms the basis of sharded (scalable) proof of work system and the characteristic known as "causal invariance."

PoW vs PoEM

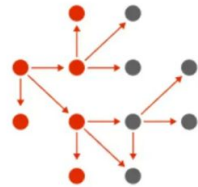
Proof-of-Work



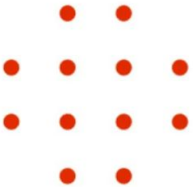
Two valid blocks are found and propagated near-simultaneously



Nodes can't discern between the two valid blocks, so they mine whichever one they saw first. This causes ~1/2 the network to mine chain RED and ~1/2 the network to mine chain BLACK.

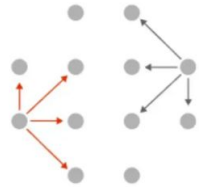


Another block is found by a miner mining on chain RED, causing the entire network to now prefer chain RED to chain BLACK.

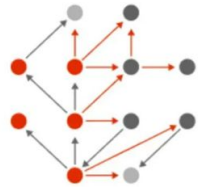


ALL NODES BACK IN CONSENSUS

Proof-of-Entropy-Minima



2 valid blocks are found and propagated near-simultaneously.



All nodes will prefer one chain extension (in this case, chain RED) to any others (in this case, chain BLACK) as soon as they are aware of it.

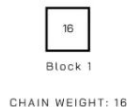


ALL NODES BACK IN CONSENSUS

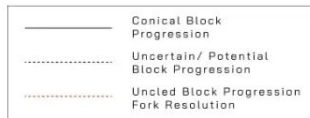
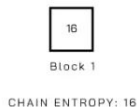
● Node
● Node mining Black
● Node mining Red

\$Qi: Tokenized Entropy

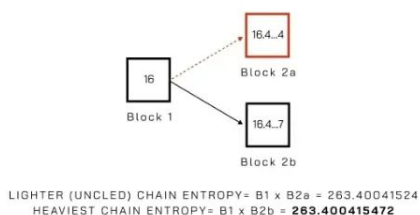
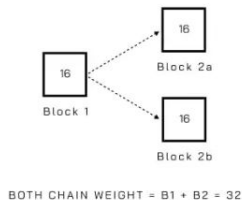
Proof-of-Work



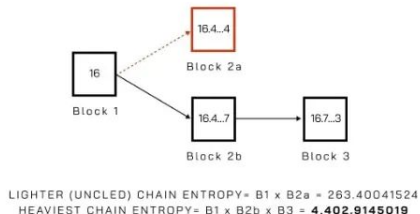
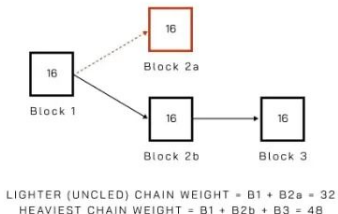
Proof-of-Entropy-Minima



The example starts with a single block of weight 16.

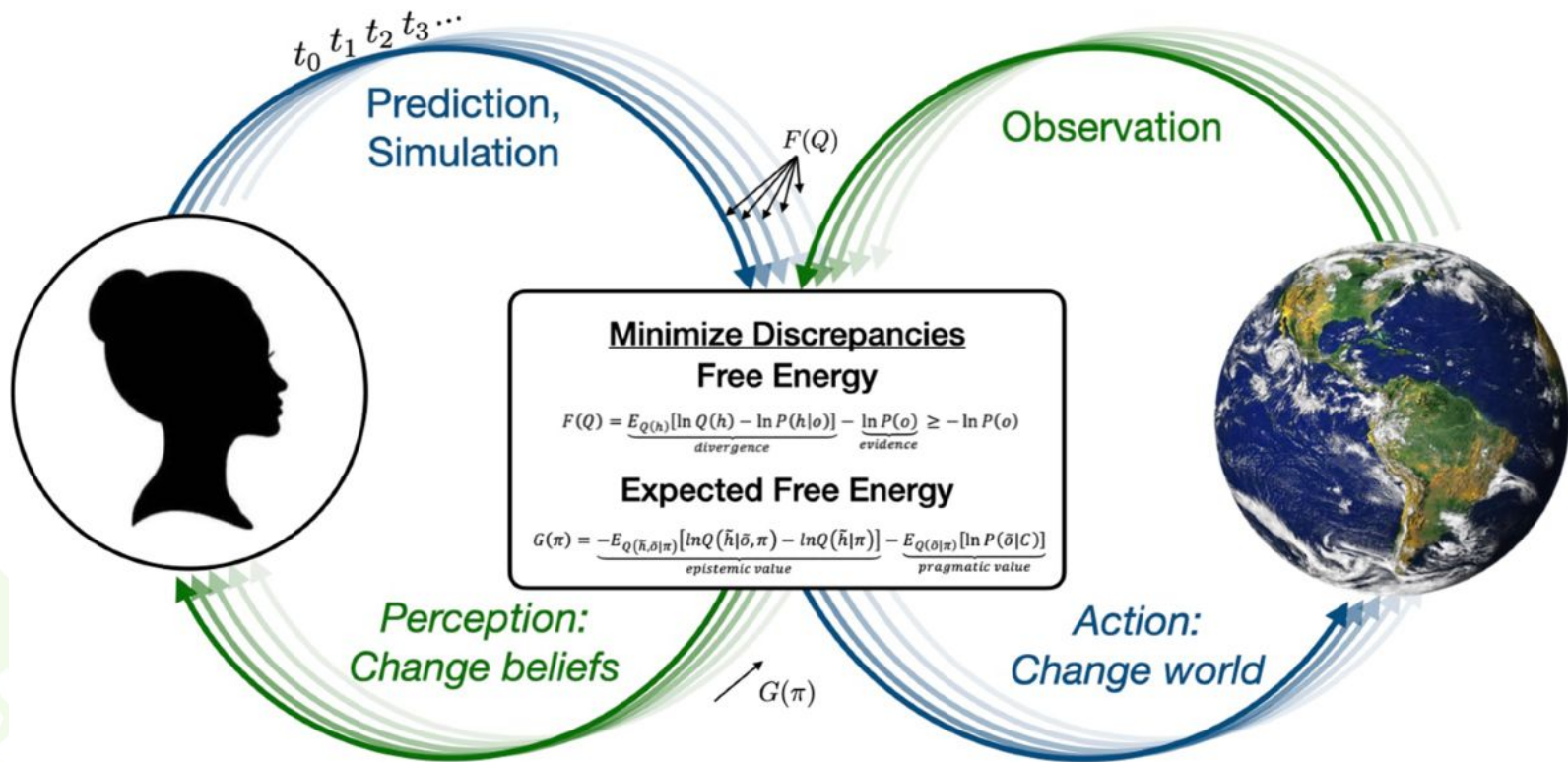


Two valid blocks are proposed simultaneously. In Proof-of-Work, this causes a fork as nodes have no preference between two valid blocks. In Proof-of-Entropy-Minima, since each block is assigned a unique weight, nodes are instantly able to prefer the chain extension that removes more entropy.



In Proof-of-Work, a third block must be found on top of one of the chain extensions in order to resolve the fork that occurred in block 2. Under Proof-of-Entropy-Minima, all nodes and miners are already working together to extend the same canonical chain.

Active Inference



Active Inference

*Belief
in the
System*

B

BLOCK
DIFFICULTY



QI REWARD

$k_{Q_i} B$

*Belief
in Belief
in the
System*

NO

YES

QI

QUAI

+

-

$$k_{Quai}(i) = k_{Quai}(i - 1)[1 \pm r]$$



QUAI REWARD

$$40 \times k_{Quai}$$

Relationship between Shannon entropy and Fisher metric

Connecting Shannon's entropy and Fisher's information

$$S_W(t) + \frac{1}{2} \ln(I_W(t)) = \frac{1}{2} \ln(2\pi e)$$

The Complexity combines Shannon's entropy and Fisher's information

$$C_W(t) = \frac{e^{2S_W(t)} I_W(t)}{\ln(2\pi e)} = 1$$

In general, the complexity measure is defined as follows

$$C = \frac{e^{2S} I}{\ln(2\pi e)} \geq 1$$

The Cramér–Rao bound is fulfilled for the Wiener process

$$I_W(t) \Delta x(t)^2 = 1 \quad \text{with} \quad \langle x(t)^2 \rangle = \Delta x(t)^2$$

We propose a generic Cramér–Rao lower bound to be used in the TE:

$$I(t) \Delta x(t)^2 \geq 1$$

The Cramér–Rao inequality is a fundamental tool in information analysis.

Proof of Entropy Minima

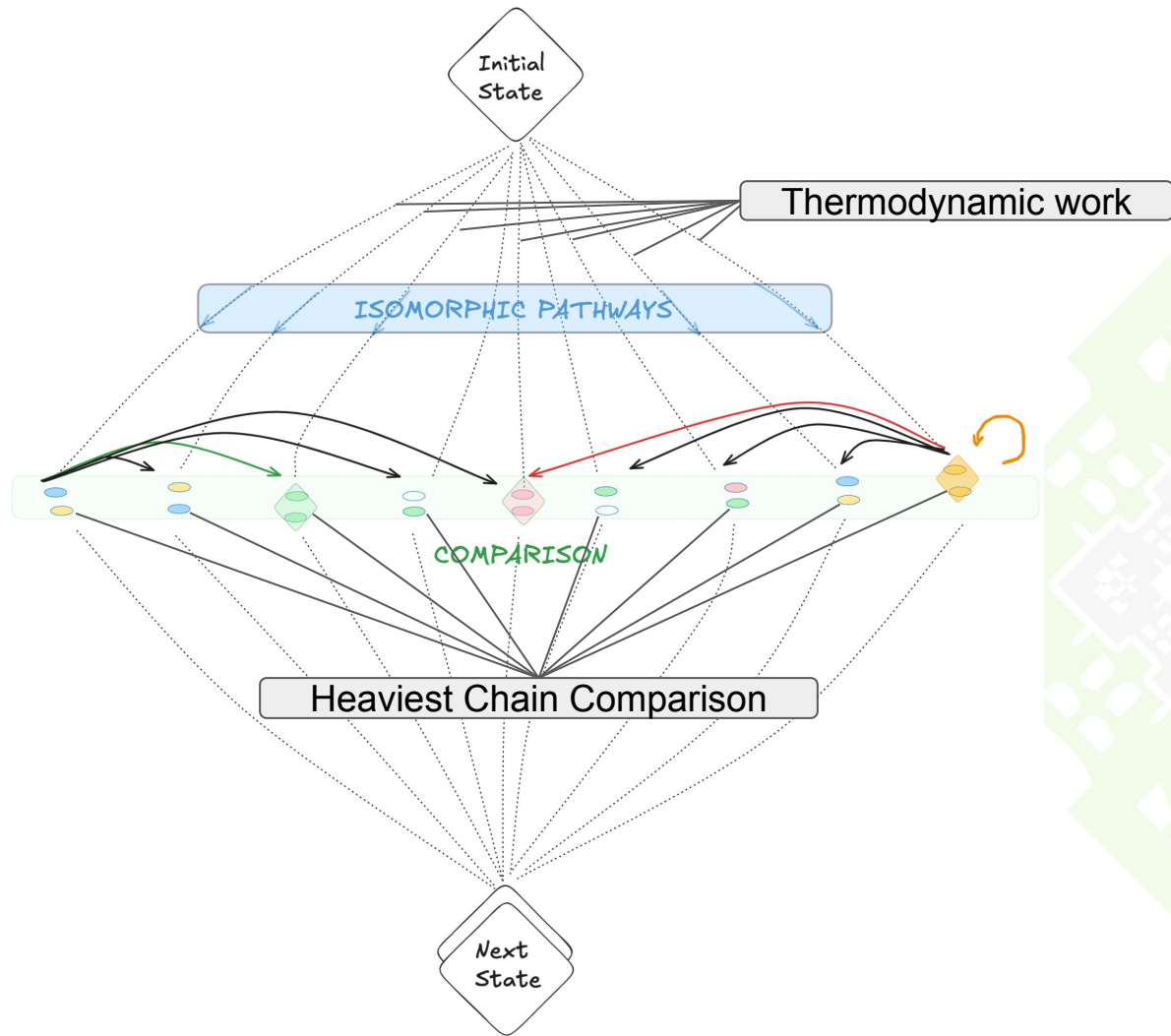
$$\triangle S = 2^{-n}$$

where n = bits removed from total possible state and $n = l - c$ where l is total states available

n	2^n	$2^{-n} = 1/(2^n)$
0	1	1
1	2	0.5
2	4	0.25
3	8	0.125
4	16	0.0625

The realized entropy of the output is c bits and the reduction in the entropy is $l - c$ bits which represent the number of leading zeros that will be called n . Additionally, n can include fractional zero bits which are after the first non-zero bit. More precisely, this makes $n = l - \log_2(\text{dint})$. The intrinsic difficulty can be used to calculate the difference entropy: $\Delta S = 1/2^n$ where ΔS represents the number of possible states removed from the macrostate in the achievement of a single block. This can be extended to compute the change in entropy in an arbitrary sequence of k blocks.

Causal Invariance



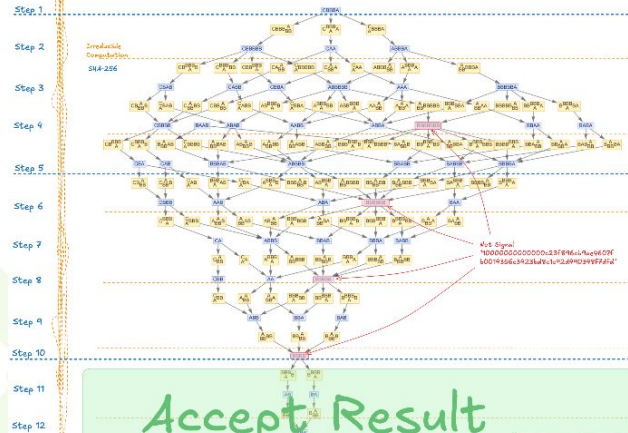
Causal Invariance

Causal Invariance

Block N

Miner A (+)

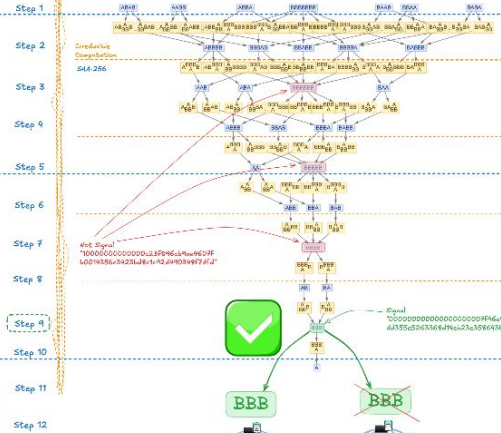
Starting Condition: 'BBB' $\rightarrow$ 'M', 'M' $\rightarrow$ 'BB', 'C' $\rightarrow$ 'M'
Signal = Minimum Difficulty Threshold: BBBs (three 'B's or less)
Starting conditions: 'CBBA', 'ABAB', 'BAAB', 'ABBA', 'BBAA', 'BABA', 'AABB', 'BBBBBB'



Block N

Miner B

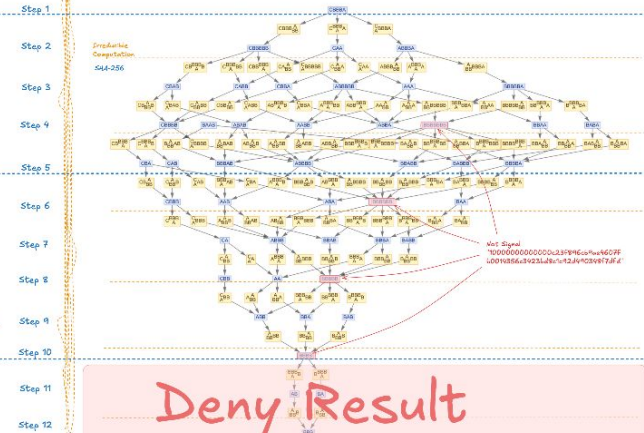
Starting Condition: 'BBB' $\rightarrow$ 'M', 'M' $\rightarrow$ 'BB'
Signal = Minimum Difficulty Threshold: BBBs (three 'B's or less)
Starting conditions: 'ABAB', 'AABB', 'ABBA', 'BBBBBB', 'BAAB', 'BBAA', 'BABA'



Block N

Miner A (-)

Starting Condition: 'BBB' $\rightarrow$ 'M', 'M' $\rightarrow$ 'BB', 'C' $\rightarrow$ 'M'
Signal = Minimum Difficulty Threshold: BBBs (three 'B's or less)
Starting conditions: 'CBBA', 'ABAB', 'BAAB', 'ABBA', 'BBAA', 'BABA', 'AABB', 'BBBBBB'

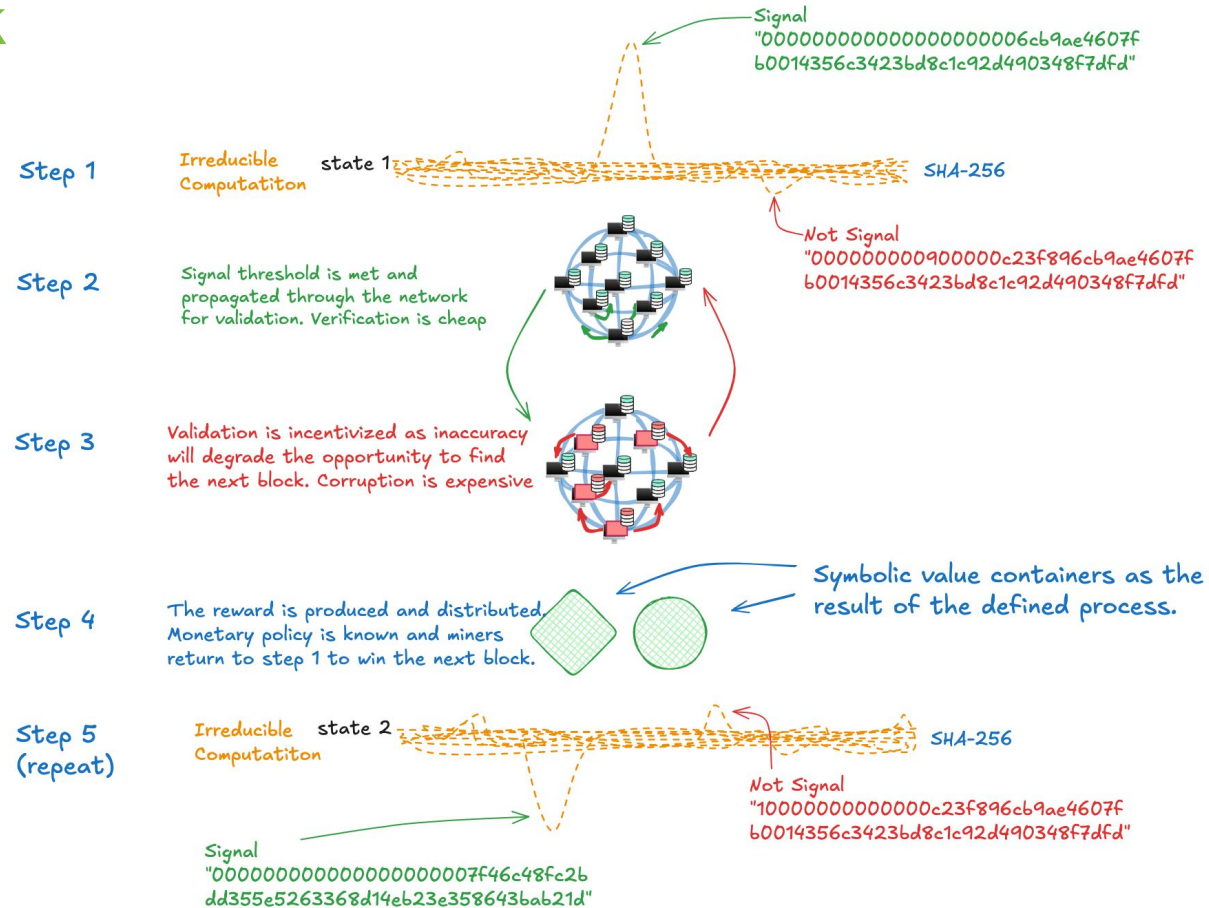


Accept Result
Move to Block $N+1$

Deny Result
Remain at Block N

PROOF OF WORK

Proof of Work



Proof of Work

Step 1

Step 2

Step 3

Step 4

Step 5

Step 6

Step 7

Step 8

Step 9

Step 10

Step 11

Step 12

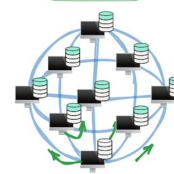
Irreducible
Computation
SHA-256

Not Signal
"10000000000000c23f896cb9ae4607f
b0014356c3423bd8c1c92d490348f7dfd"

Signal
"00000000000000000000007f46c48fc2b
dd355e5263368d14eb23e358643bab21d"



BBB



~~BBB~~

